

Q1: A helicopter is flying horizontally with a speed 'v' at an altitude 'h' has to drop a food packet for a man on the ground. What is the distance of helicopter from the man when the food packet is dropped?

(A) $\sqrt{\frac{2ghv^2+1}{h^2}}$

(B) $\sqrt{2ghv^2 + h^2}$

(C) $\sqrt{\frac{2v^2h}{g} + h^2}$

(D) $\sqrt{\frac{2gh}{v^2} + h^2}$

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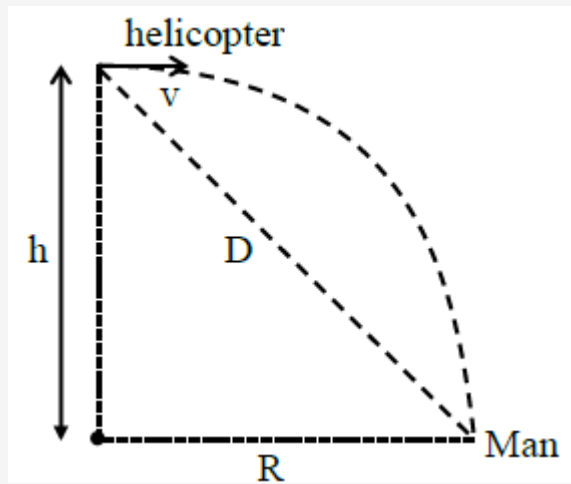
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Solution:



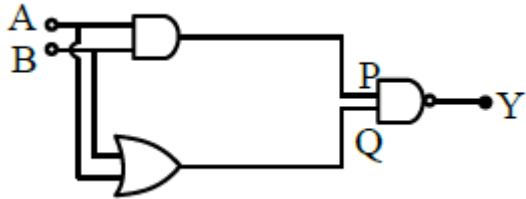
$$R = \sqrt{\frac{2h}{g}} \cdot v$$

$$D = \sqrt{R^2 + h^2}$$

$$= \sqrt{\left(\sqrt{\frac{2h}{g}} \cdot v \right)^2 + h^2}$$

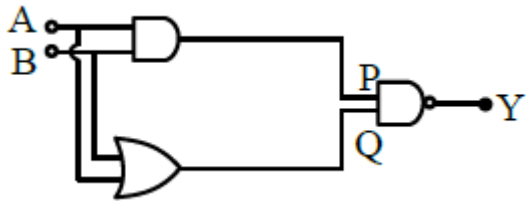
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Q2: In the following logic circuit the sequence of the inputs A, B are (0, 0), (0, 1), (1, 0) and (1, 1). The output Y for this sequence will be :



- (A) 1, 0, 1, 0
- (B) 0, 1, 0, 1
- (C) 1, 1, 1, 0
- (D) 0, 0, 1, 1

Q2: In the following logic circuit the sequence of the inputs A, B are (0, 0), (0, 1), (1, 0) and (1, 1). The output Y for this sequence will be :



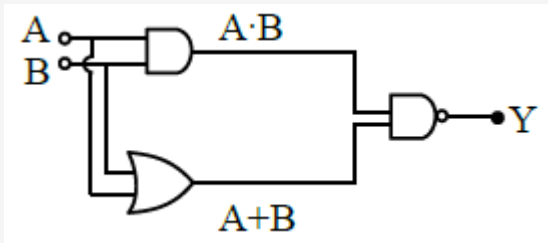
(A) 1, 0, 1, 0

(B) 0, 1, 0, 1

(C) 1, 1, 1, 0

(D) 0, 0, 1, 1

Solution:



$$Y = \overline{(A \cdot B) \cdot (A + B)}$$

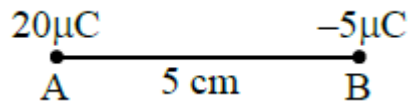
$$Y_{(0,0)} = 1$$

$$Y_{(0,1)} = 1$$

$$Y_{(1,0)} = 1$$

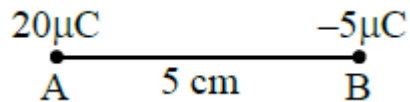
$$Y_{(1,1)} = 0$$

Q3: Two particles A and B having charges $20\ \mu\text{C}$ and $-5\ \mu\text{C}$ respectively are held fixed with a separation of 5 cm. At what position a third charged particle should be placed so that it does not experience a net electric force?



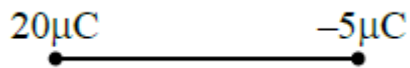
- (A) At 5 cm from $20\ \mu\text{C}$ on the left side of system
- (B) At 5 cm from $-5\ \mu\text{C}$ on the right side
- (C) At 1.25 cm from $-5\ \mu\text{C}$ between two charges
- (D) At midpoint between two charges

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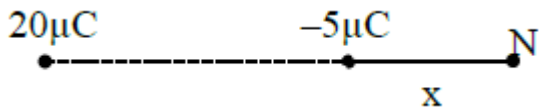


- (A) At 5 cm from $20\ \mu\text{C}$ on the left side of system
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- (C) At 1.25 cm from $-5\ \mu\text{C}$ between two charges
- (D) At midpoint between two charges

Solution:



Null point is possible only right side of $-5\ \mu\text{C}$



$$E_N = +\frac{k(-5\ \mu\text{C})}{x^2} + \frac{k(20\ \mu\text{C})}{(5+x)^2} = 0$$

$$x = 5\ \text{cm}$$

Q4: A reversible engine has an efficiency of $\frac{1}{4}$. If the temperature of the sink is reduced by 58°C , its efficiency becomes double. Calculate the temperature of the sink :

- (A) 174 K
- (B) 280 K
- (C) 180.4 K
- (D) 382 K

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(D) 382 K

Solution:

T_2 = sink temperature

$$\eta = 1 - \frac{T_2}{T_1}$$

$$\frac{1}{4} = 1 - \frac{T_2}{T_1}$$

$$\frac{T_2}{T_1} = \frac{3}{4} \dots (i)$$

$$\frac{1}{2} = 1 - \frac{T_2 - 58}{T_1}$$

$$\frac{T_2}{T_1} - \frac{58}{T_1} = \frac{1}{2}$$

$$\frac{3}{4} = \frac{58}{T_1} + \frac{1}{2}$$

$$\frac{1}{4} = \frac{58}{T_1} \Rightarrow T_1 = 232$$

$$T_2 = \frac{3}{4} \times 232$$

$$T_2 = 174 \text{ K}$$

Q5: An object is placed at the focus of concave lens having focal length f . What is the magnification and distance of the image from the optical centre of the lens?

- (A) $1, \infty$
- (B) Very high, ∞
- (C) $\frac{1}{2}, \frac{f}{2}$
- (D) $\frac{1}{4}, \frac{f}{4}$

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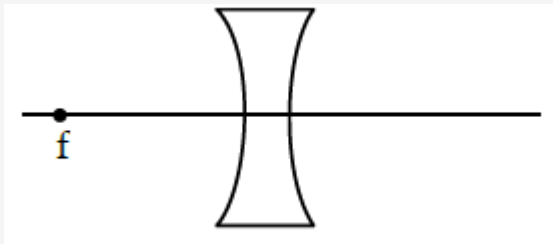
(A) $1, \infty$

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Solution:



$$U = -f$$

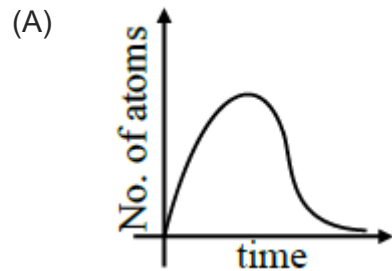
$$\frac{1}{V} - \frac{1}{U} = \frac{1}{-f} \Rightarrow \frac{1}{V} = -\frac{2}{f}$$

$$V = \frac{-f}{2}$$

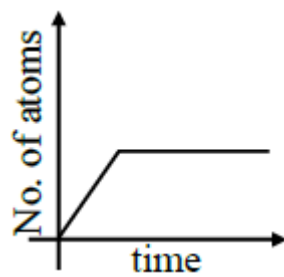
$$m = \frac{V}{U} = \frac{1}{2}$$

$$\text{distance} = \frac{f}{2}$$

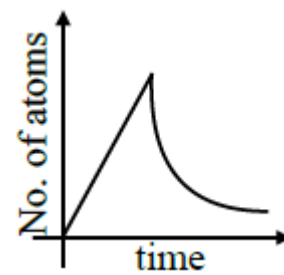
Q6: A sample of a radioactive nucleus A disintegrates to another radioactive nucleus B, which in turn disintegrates to some other stable nucleus C. Plot of a graph showing the variation of number of atoms of nucleus B versus time is :
(Assume that at $t = 0$, there are no B atoms in the sample)



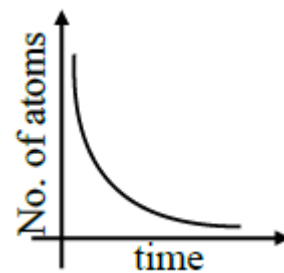
(B)



(C)

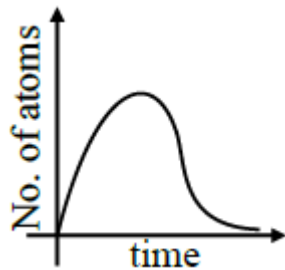


(D)

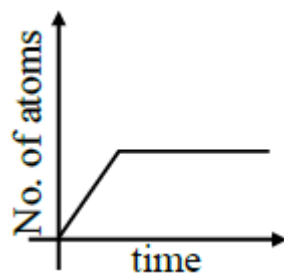


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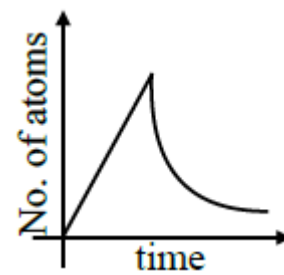
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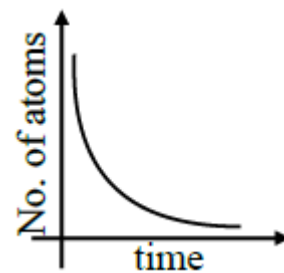
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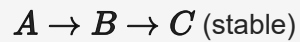
(C)



(D)



Solution:



Initially no. of atoms of B = 0 after $t = 0$, no. of atoms of B will start increasing & reaches maximum value when rate of decay of B = rate of formation of B.

After that maximum value, no. of atoms will start decreasing as growth & decay both are exponential functions, so best possible graph is (2)

Q7: A coil having N turns is wound tightly in the form of a spiral with inner and outer radii 'a' and 'b' respectively. Find the magnetic field at centre, when a current I passes through coil :

(A) $\frac{\mu_0 I N}{2(b-a)} \log_e \left(\frac{b}{a} \right)$

(B) $\frac{\mu_0 I}{8} \left[\frac{a+b}{a-b} \right]$

(C) $\frac{\mu_0 I}{4(a-b)} \left[\frac{1}{a} - \frac{1}{b} \right]$

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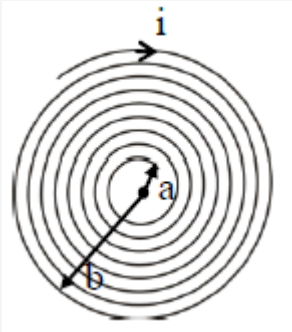
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Solution:



No. of turns in dx width = $\frac{N}{b-a} dx$

$$\int dB = \int_a^b \left(\frac{N}{b-a} \right) dx \frac{\mu_0 i}{2x}$$

$$B = \frac{N\mu_0 i}{2(b-a)} \ln \left(\frac{b}{a} \right)$$

Q8: A body of mass M moving at speed V_0 collides elastically with a mass ' m ' at rest. After the collision, the two masses move at angle θ_1 and θ_2 with respect to the initial direction of motion of the body of mass M . The largest possible value of the ratio M/m , for which the angles θ_1 and θ_2 will be equal, is :

- (A) 4
- (B) 1
- (C) 3
- (D) 2

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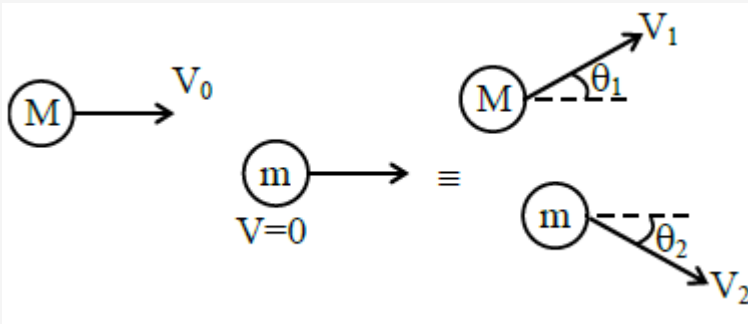
(A) 4

(B) 1

(C) 3

(D) 2

Solution:



given $\theta_1 = \theta_2 = \theta$

from momentum conservation

in x-direction $MV_0 = MV_1 \cos \theta + mV_2 \cos \theta$

in y-direction $0 = MV_1 \sin \theta - mV_2 \sin \theta$

Solving above equations

$$V_2 = \frac{MV_1}{m}, V_0 = 2V_1 \cos \theta$$

From energy conservation

$$\frac{1}{2}MV_0^2 = \frac{1}{2}MV_1^2 + \frac{1}{2}mV_2^2$$

Substituting value of V_2 & V_0 , we will get

$$\frac{M}{m} + 1 = 4\cos^2 \theta \leq 4$$

$$\frac{M}{m} \leq 3$$

Q9: The masses and radii of the earth and moon are (M_1, R_1) and (M_2, R_2) respectively. Their centres are at a distance 'r' apart. Find the minimum escape velocity for a particle of mass 'm' to be projected from the middle of these two masses:

(A) $V = \frac{1}{2} \sqrt{\frac{4G(M_1+M_2)}{r}}$

(B) $V = \sqrt{\frac{4G(M_1+M_2)}{r}}$

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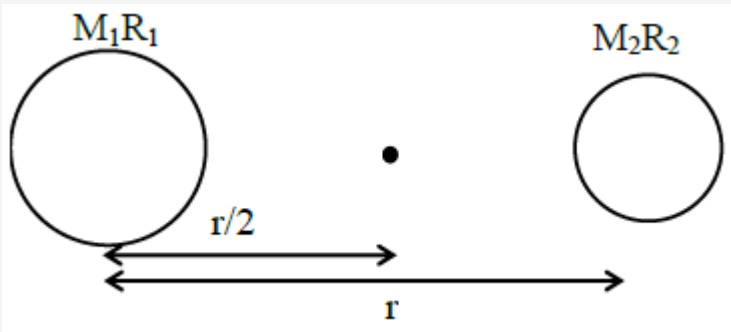
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Solution:



$$\frac{1}{2}mV^2 - \frac{GM_1m}{r/2} - \frac{GM_2m}{r/2} = 0$$

$$\frac{1}{2}mV^2 = \frac{2Gm}{r}(M_1 + M_2)$$

$$V = \sqrt{\frac{4G(M_1 + M_2)}{r}}$$

Q10: A small square loop of side 'a' and one turn is placed inside a larger square loop of side b and one turn ($b \gg a$). The two loops are coplanar with their centres coinciding. If a current I is passed in the square loop of side 'b', then the coefficient of mutual inductance between the loops is :

(A) $\frac{\mu_0}{4\pi} 8\sqrt{2} \frac{a^2}{b}$

(B) $\frac{\mu_0}{4\pi} \frac{8\sqrt{2}}{a}$

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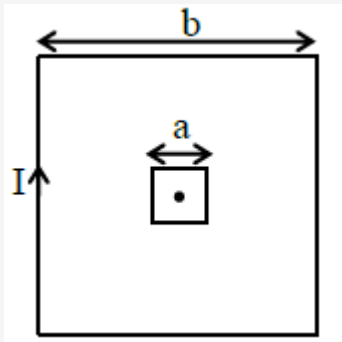
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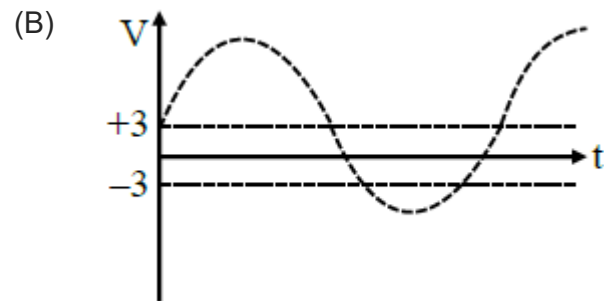
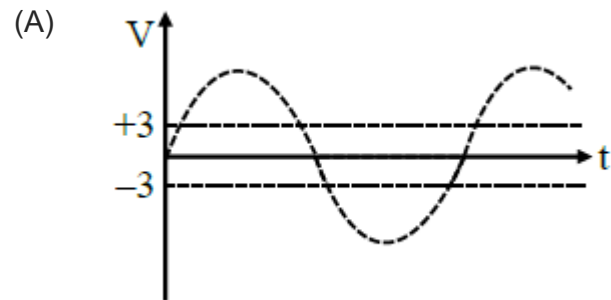
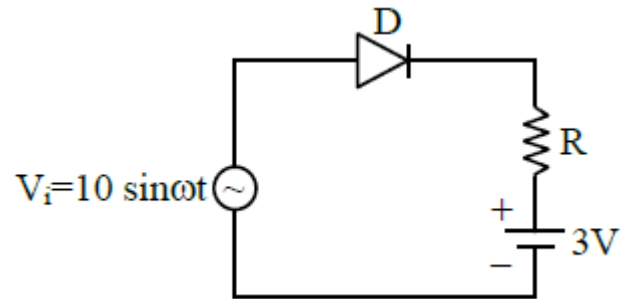


$$B = \left[\frac{\mu_0}{4\pi} \frac{I}{b/2} \times 2 \sin 45 \right] \times 4$$

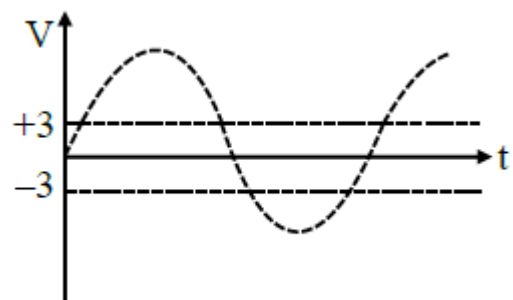
$$\phi = 2\sqrt{2} \frac{\mu_0}{\pi} \frac{I}{b} \times a^2$$

$$\therefore M = \frac{\phi}{I} = \frac{2\sqrt{2}\mu_0 a^2}{\pi b} = \frac{\mu_0}{4\pi} 8\sqrt{2} \frac{a^2}{b}$$

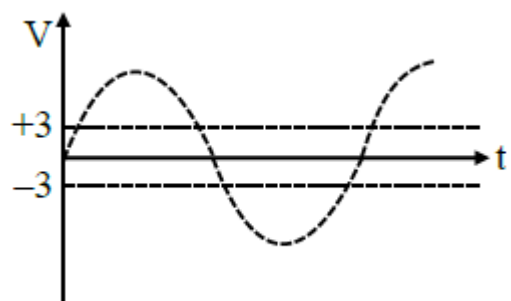
Q11: Choose the correct waveform that can represent the voltage across R of the following circuit, assuming the diode is ideal one:



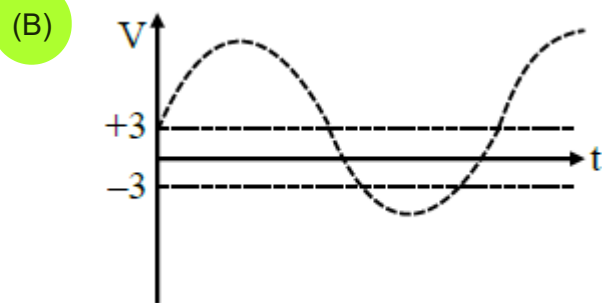
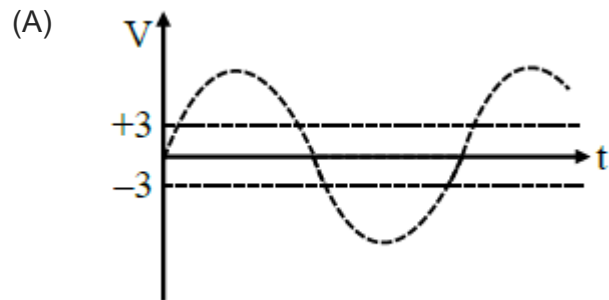
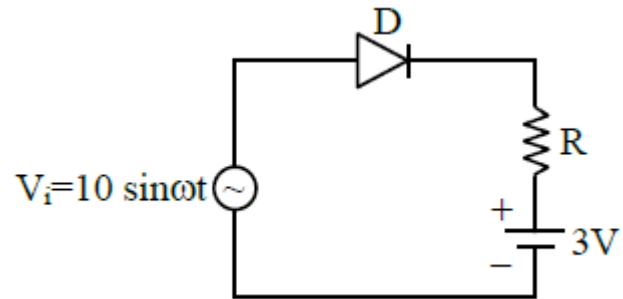
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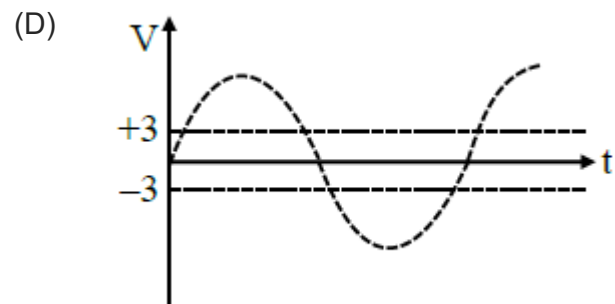
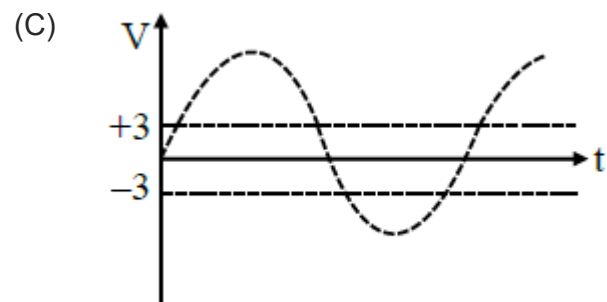


(D)



Q11: Choose the correct waveform that can represent the voltage across R of the following circuit, assuming the diode is ideal one:





Solution:

When $V_i > 3$ volt, $V_R > 0$

Because diode will be in forward biased state

When $V_i \leq 3$ volt; $V_R = 0$

Because diode will be in reverse biased state.

Q12: A uniform heavy rod of weight 10 kg ms^{-2} , cross-sectional area 100 cm^2 and length 20 cm is hanging from a fixed support. Young modulus of the material of the rod is $2 \times 10^{11} \text{ Nm}^{-2}$. Neglecting the lateral contraction, find the elongation of rod due to its own weight.

(A) $2 \times 10^{-9} \text{ m}$

(B) $5 \times 10^{-8} \text{ m}$

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(D) $5 \times 10^{-10} \text{ m}$

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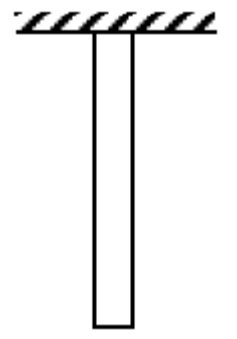
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Solution:



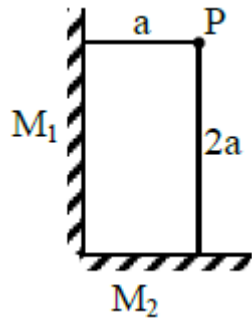
We know,

$$\Delta \ell = \frac{WL}{2AY}$$

$$\Delta \ell = \frac{10 \times 1}{2 \times 5} \times 100 \times 10^{-4} \times 2 \times 10^{11}$$

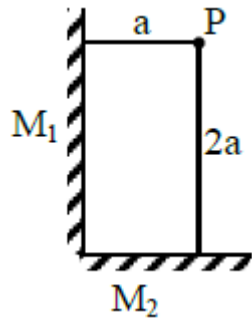
$$\Delta \ell = \frac{1}{2} \times 10^{-9} = 5 \times 10^{-10} m$$

Q13: Two plane mirrors M_1 and M_2 are at right angle to each other shown. A point source 'P' is placed at 'a' and '2a' meter away from M_1 and M_2 respectively. The shortest distance between the images thus formed is : (Take $\sqrt{5} = 2.3$)



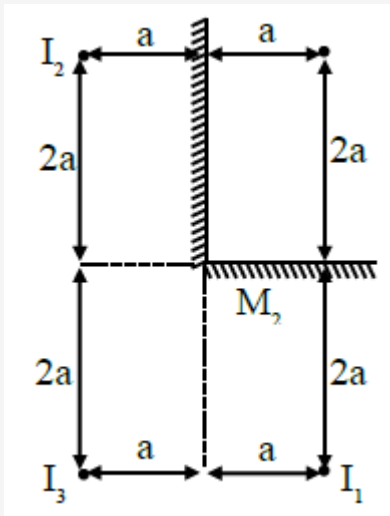
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- (C) $2.3 a$
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Solution:



Shortest distance is $2a$ between I_1 & I_3

But answer given is for I_1 & I_2

$$= \sqrt{(4a)^2 + (2a)^2}$$

$$= a\sqrt{20}$$

$$= 4.47 a$$

Q14: Match **List-I** with **List-II**.

List-I	List-II
(a) Torque	(i) MLT^{-1}
(b) Impulse	(ii) MT^{-2}
(c) Tension	(iii) ML^2T^{-2}
(d) Surface Tension	(iv) MLT^{-2}

Choose the **most appropriate** answer from the option given below :

- (A) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (B) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
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(C) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(D) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

Solution:

torque $\tau \rightarrow ML^2T^{-2}$ (III)

Impulse $I \Rightarrow MLT^{-1}$ (I)

Tension force $\Rightarrow MLT^{-2}$ (IV)

Surface tension $\Rightarrow MT^{-2}$ (II)

Q15: For an ideal gas the instantaneous change in pressure 'p' with volume 'v' is given by the equation $\frac{dp}{dv} = -ap$. If $p = p_0$ at $v = 0$ is the given boundary condition, then the maximum temperature one mole of gas can attain is:

(Here R is the gas constant)

(A) $\frac{p_0}{a e R}$

(B) $\frac{ap_0}{e R}$

(C) infinity

(D) $0^\circ C$

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Solution:

$$\int_{p_0}^p \frac{dp}{P} = -a \int_0^v dv$$

$$\ln \left(\frac{p}{p_0} \right) = -av$$

$$p = p_0 e^{-av}$$

For temperature maximum p-v product should be maximum

$$T = \frac{pv}{nR} = \frac{p_0 v e^{-av}}{R}$$

$$\frac{dT}{dv} = 0 \Rightarrow \frac{p_0}{R} \{ e^{-av} + v e^{-av} (-a) \}$$

$$\frac{p_0 e^{-av}}{R} \{ 1 - av \} = 0$$

$$v = \frac{1}{a}, \infty$$

$$T = \frac{p_0 \times 1}{Rae} = \frac{p_0}{Rae}$$

at $v = \infty$

$$T = 0$$

Q16: Which of the following equations is dimensionally incorrect?

Where t = time, h = height, s = surface tension, θ = angle, ρ = density, a, r = radius, g = acceleration due to gravity, v = volume, p = pressure, W = work done, Γ = torque, ϵ = permittivity, E = electric field, J = current density, L = length.

(A) $v = \frac{\pi p a^4}{8 \eta L}$

(B) $h = \frac{2s \cos \theta}{\rho r g}$

(C) $J = \epsilon \frac{\partial E}{\partial t}$

(D) $W = \Gamma \theta$

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Solution:

(i) $\frac{\pi p a^4}{8 \eta L} = \frac{dv}{dt}$ = Volumetric flow rate (poiseuille's law)

(ii) $h \rho g = \frac{2s}{r} \cos \theta$

(iii) $RHS \Rightarrow \epsilon \times \frac{1}{4 \pi \epsilon_0} \frac{a}{r^2} \times \frac{1}{\epsilon} = \frac{q}{t} \times \frac{1}{r^2}$
 $= \frac{I}{L^2} = IL^{-2}$

$LHS \Rightarrow T = \frac{I}{A} = IL^{-2}$

(iv) $W = \tau \theta$

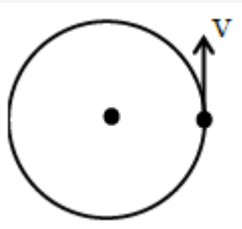
Q17: Angular momentum of a single particle moving with constant speed along circular path :

- (A) changes in magnitude but remains same in the direction
- (B) remains same in magnitude and direction
- (C) remains same in magnitude but changes in the direction
- (D) is zero

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- (B) remains same in magnitude and direction
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- (D) is zero

Solution:



$$|\vec{L}| = mvr$$

And direction will be upward & remain constant

Q18: In an ac circuit, an inductor, a capacitor and a resistor are connected in series with $X_L = R = X_C$. Impedance of this circuit is :

- (A) $2R^2$
- (B) Zero
- (C) R
- (D) $R\sqrt{2}$

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- (B) Zero
- (C) R
- (D) $R\sqrt{2}$

Solution:

$$Z = \sqrt{(X_L - X_C)^2 + R^2} = R \quad \because X_L = X_C$$

Q19: A moving proton and electron have the same de-Broglie wavelength. If K and P denote the K.E. and momentum respectively. Then choose the correct option :

- (A) $K_p < K_e$ and $P_p = P_e$
- (B) $K_p = K_e$ and $P_p = P_e$
- (C) $K_p < K_e$ and $P_p < P_e$
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(D) $K_p > K_e$ and $P_p = P_e$

Solution:

$$\lambda_P = \frac{h}{P_P} \quad \lambda_e = \frac{h}{P_e}$$

$$\therefore \lambda_P = \lambda_e$$

$$\Rightarrow P_P = P_e$$

$$(K)_P = \frac{P_P^2}{2m_P}$$

$$(K)_e = \frac{P_e^2}{2m_e}$$

$$K_P < K_e \text{ as } m_P > m_e$$

Q20: Consider a galvanometer shunted with 5Ω resistance and 2% of current passes through it.

What is the resistance of the given galvanometer?

- (A) $300\ \Omega$
- (B) $344\ \Omega$
- (C) $245\ \Omega$
- (D) $226\ \Omega$

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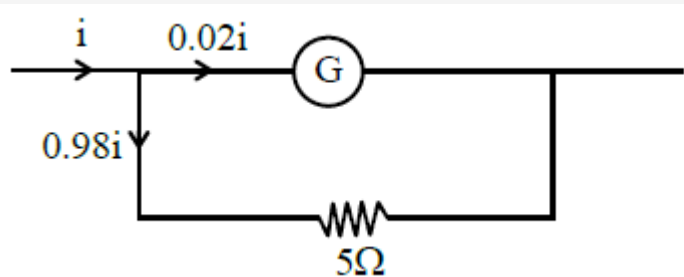
(A) $300\ \Omega$

(B) $344\ \Omega$

(C) $245\ \Omega$

(D) $226\ \Omega$

Solution:



$$0.02i \times R_g = 0.98i \times 5$$

$$R_g = 245\ \Omega$$

Q21: When a rubber ball is taken to a depth of _____m in deep sea, its volume decreases by 0.5%.

(The bulk modulus of rubber = $9.8 \times 10^8 \text{ Nm}^{-2}$ Density of sea water = 10^3 kgm^{-3}

$g = 9.8 \text{ m/s}^2$)

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$g = 9.8 \text{ m/s}^2$)

500.00

Solution:

$$B = -\frac{\Delta P}{\left(\frac{\Delta V}{V}\right)} = -\frac{\rho gh}{\left(\frac{\Delta V}{V}\right)}$$

$$-\frac{B \frac{\Delta V}{V}}{\rho g} = h$$

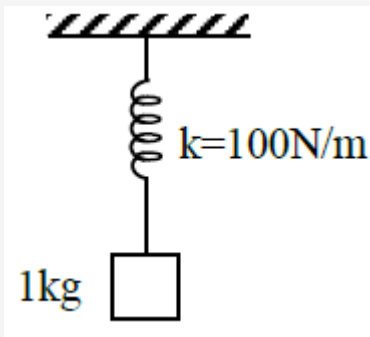
$$\frac{9.8 \times 10^8 \times 0.5}{100 \times 10^3 \times 9.8} = h$$

$$h = 500 \text{ m}$$

Q22: A particle of mass 1 kg is hanging from a spring of force constant 100 Nm^{-1} . The mass is pulled slightly downward and released so that it executes free simple harmonic motion with time period T. The time when the kinetic energy and potential energy of the system will become equal, is $\frac{T}{x}$. The value of x is _____ .

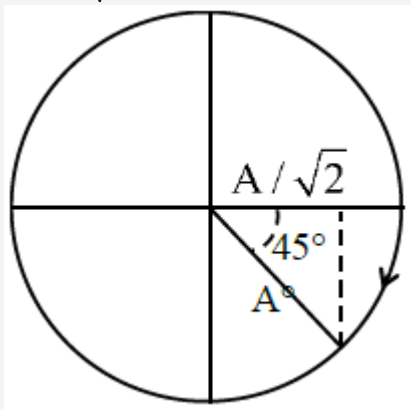
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Solution:



$$KE = PE$$

$$y = \frac{A}{\sqrt{2}} = A \sin \omega t$$



$$t = \frac{T}{8} = \frac{T}{x}$$

$$x = 8$$

Q23: If the sum of the heights of transmitting and receiving antennas in the line of sight of communication is fixed at 160 m, then the maximum range of LOS communication is _____ km.

(Take radius of Earth = 6400 km)

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(Take radius of Earth = 6400 km)

Solution:

$$h_T = h_R = 160 \dots (i)$$

$$d = \sqrt{2RH_T} + \sqrt{2RH_R}$$

$$d = \sqrt{2R} [\sqrt{h_T} + \sqrt{h_R}]$$

$$d = \sqrt{2R} [\sqrt{x} + \sqrt{160 - x}]$$

$$\frac{d(d)}{dx} = 0$$

$$\frac{1}{2\sqrt{x}} + \frac{1(-1)}{2\sqrt{160 - x}} = 0$$

$$\frac{1}{\sqrt{x}} = \frac{1}{\sqrt{160 - x}}$$

$$x = 80 \text{ m}$$

$$d_{\max} = \sqrt{2 \times 6400} \left[\sqrt{\frac{80}{1000}} + \sqrt{\frac{20}{1000}} \right]$$

$$= \frac{80\sqrt{2} \times 2\sqrt{80}}{10\sqrt{10}}$$

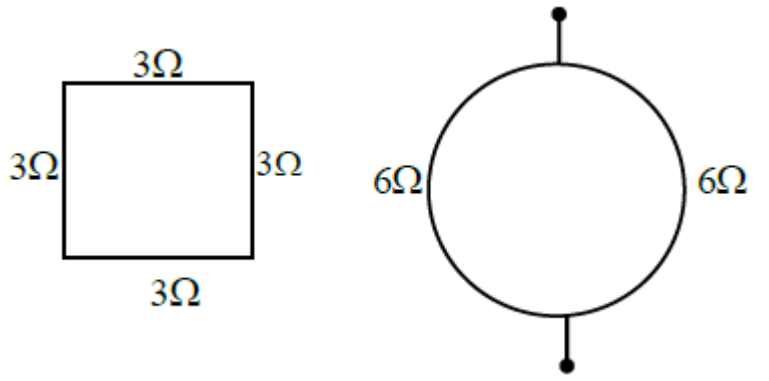
$$= 8 \times 2 \times \sqrt{2} \times 2\sqrt{2} = 64 \text{ km}$$

Q24: A square shaped wire with resistance of each side $3\ \Omega$ is bent to form a complete circle. The resistance between two diametrically opposite points of the circle in unit of Ω will be

Q24: A square shaped wire with resistance of each side 3Ω is bent to form a complete circle. The resistance between two diametrically opposite points of the circle in unit of Ω will be

3.00

Solution:



$$R_{eq} = 3\Omega$$

Q25: A wire having a linear mass density $9.0 \times 10^{-4} \text{ kg/m}$ is stretched between two rigid supports with a tension of 900 N. The wire resonates at a frequency of 500 Hz. The next higher frequency at which the same wire resonates is 550 Hz. The length of the wire is _____m.

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10.00

Solution:

$$\mu = 9.0 \times 10^{-4} \frac{\text{kg}}{\text{m}}$$

$$T = 900 \text{ N}$$

$$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{900}{9 \times 10^{-4}}} = 1000 \text{ m/s}$$

$$f_1 = 500 \text{ Hz}$$

$$f = 550$$

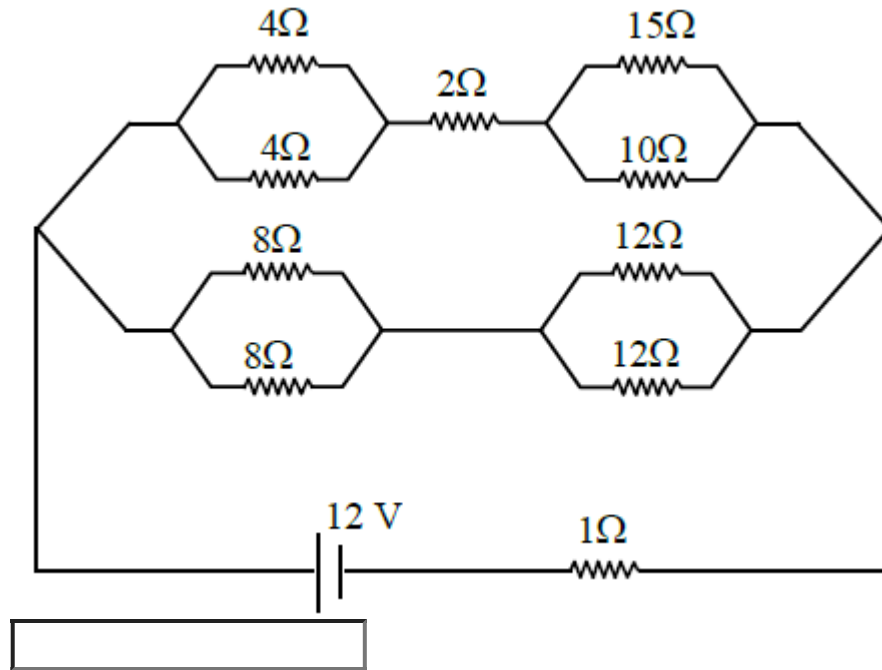
$$\frac{nV}{2\ell} = 500 \dots\dots (i)$$

$$\frac{(n+1)V}{2\ell} = 550 \dots\dots (ii)$$

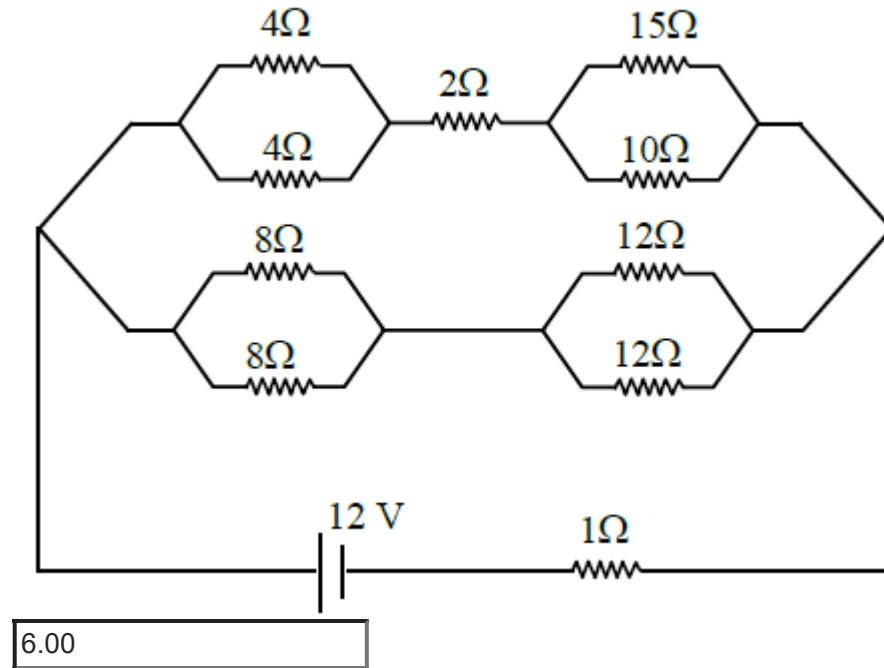
$$\text{From (ii) \& (i), } \frac{V}{2\ell} = 50$$

$$\ell = \frac{1000}{2 \times 50} = 10 \text{ m}$$

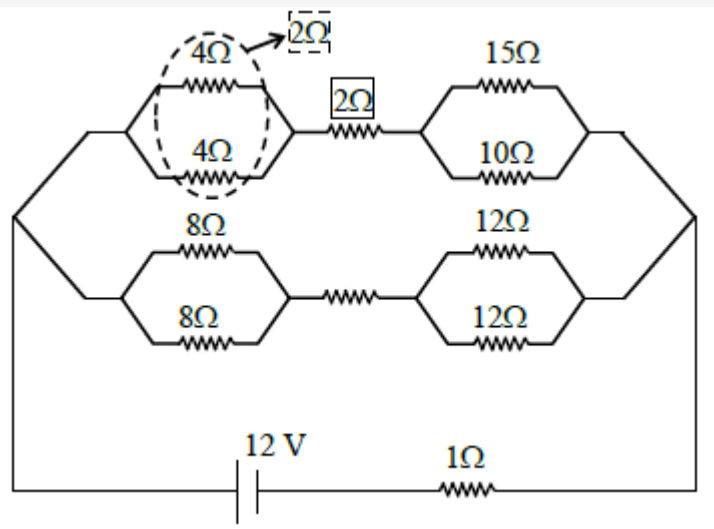
Q26: The voltage drop across $15\ \Omega$ resistance in the given figure will be _____ V.



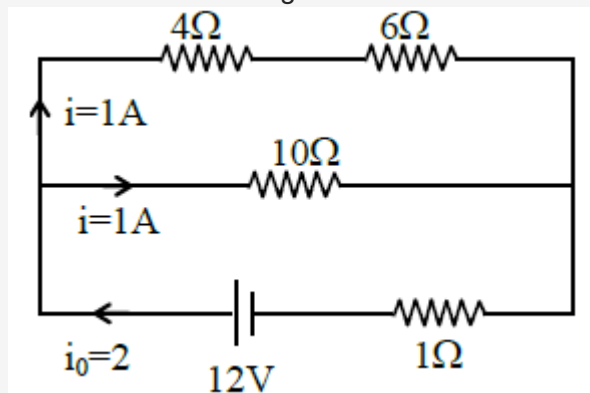
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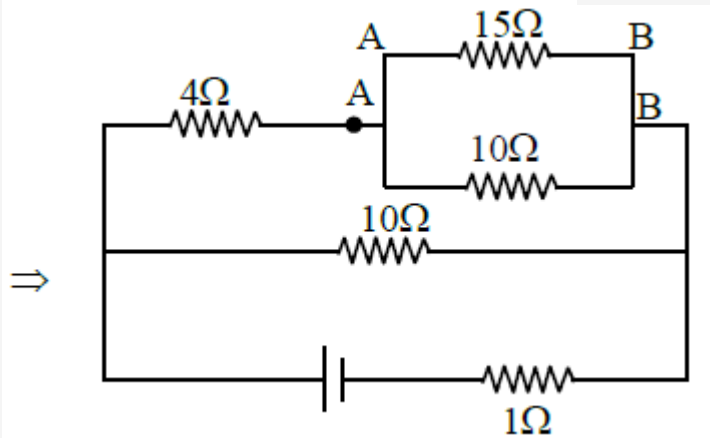
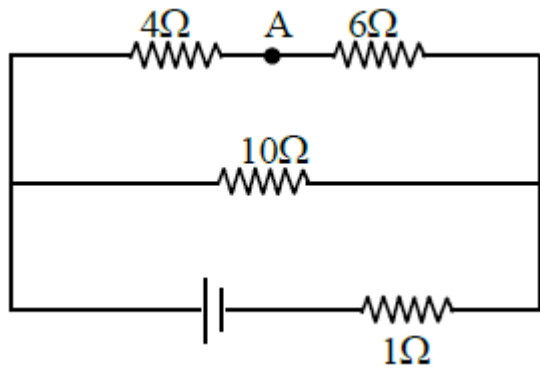


Solution:



⇒ effective circuit diagram will be





Point drop across $6\Omega = 1 \times 6 = 6 = V_{AB}$

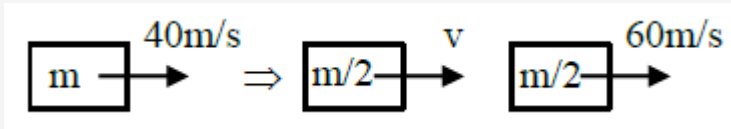
$\Rightarrow$ Hence point drop across $15\Omega = 6 \text{ volt} = V_{AB}$

Q27: A block moving horizontally on a smooth surface with a speed of 40 ms^{-1} splits into two equal parts. If one of the parts moves at 60 ms^{-1} in the same direction, then the fractional change in the kinetic energy will be $x : 4$ where $x =$ _____.

Q27: A block moving horizontally on a smooth surface with a speed of 40 ms^{-1} splits into two equal parts. If one of the parts moves at 60 ms^{-1} in the same direction, then the fractional change in the kinetic energy will be $x : 4$ where $x = \underline{\hspace{2cm}}$.

1.00

Solution:



$$P_i = P_f$$

$$m \times 40 = \frac{m}{2} \times v + \frac{m}{2} \times 60$$

$$40 = \frac{v}{2} + 30$$

$$\Rightarrow v = 20$$

$$(K.E.)_i = \frac{1}{2} m \times (40)^2 = 800m$$

$$(K.E.)_f = \frac{1}{2} \frac{m}{2} \cdot (20)^2 + \frac{1}{2} \cdot \frac{m}{2} (60)^2 = 1000 m$$

$$|\Delta K.E.| = |1000m - 800m| = 200m$$

$$\frac{\Delta K.E.}{(K.E.)_i} = \frac{200m}{800m} = \frac{1}{4} = \frac{x}{4}$$

$$x = 1$$

Q28: The electric field in an electromagnetic wave is given by

$$E = (50 \text{ NC}^{-1}) \sin \omega (t - x/c)$$

The energy contained in a cylinder of volume V is $5.5 \times 10^{-12} \text{ J}$. The value of V is _____ cm^3 .

(given $\epsilon_0 = 8.8 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$)

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(given $\epsilon_0 = 8.8 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$)

500.00

Solution:

$$E = 50 \sin\left(\omega t - \frac{\omega}{c} \cdot x\right)$$

$$\text{Energy density} = \frac{1}{2} \epsilon_0 E_0^2$$

$$\text{Energy for volume } V = \frac{1}{2} \epsilon_0 E_0^2 \cdot V = 5.5 \times 10^{-12}$$

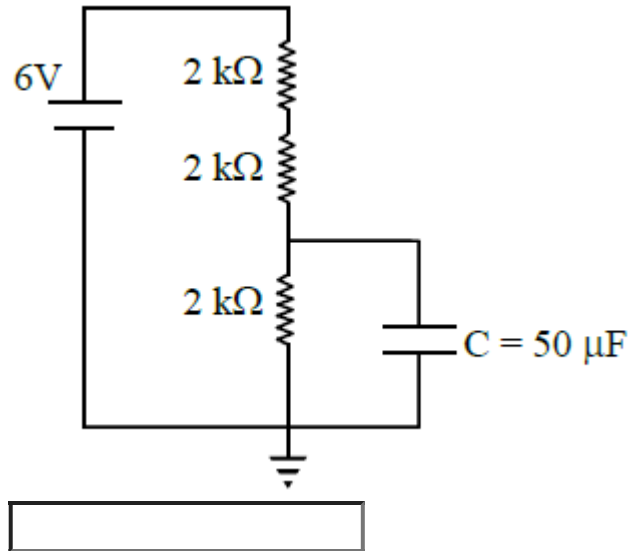
$$\frac{1}{2} 8.8 \times 10^{-12} \times 2500V = 5.5 \times 10^{-12}$$

$$V = \frac{5.5 \times 2}{2500 \times 8.8} = .0005 \text{ m}^3$$

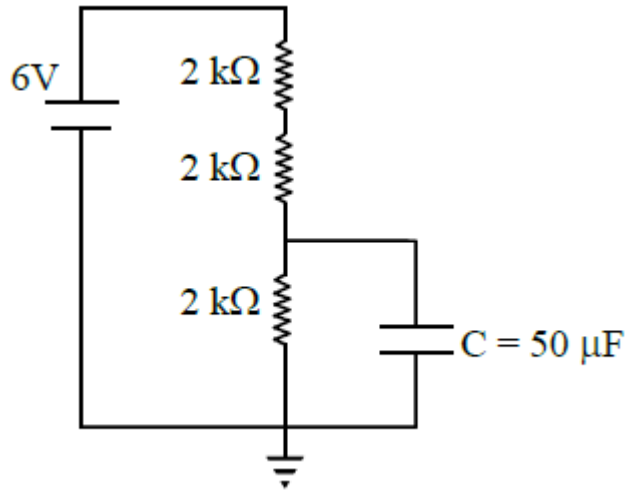
$$= .0005 \times 10^6 (\text{c.m})^3$$

$$= 500 (\text{c.m})^3$$

Q29: A capacitor of $50\ \mu\text{F}$ is connected in a circuit as shown in figure. The charge on the upper plate of the capacitor is _____ μC .

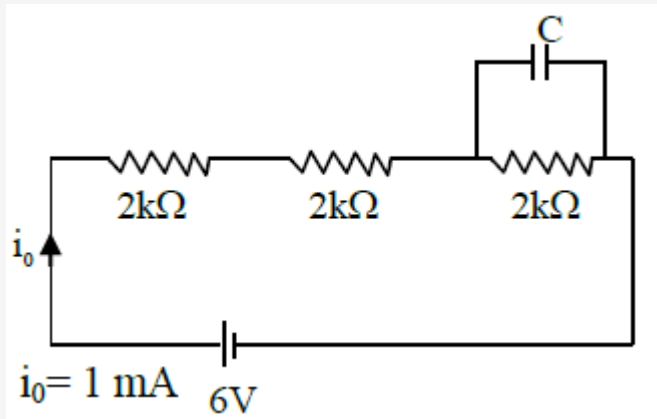


Q29: A capacitor of $50\ \mu\text{F}$ is connected in a circuit as shown in figure. The charge on the upper plate of the capacitor is _____ μC .



100.00

Solution:



Potential Difference across each resistor = $2V$

$$q = CV$$

$$= 50 \times 10^{-6} \times 2 = 100 \times 10^{-6} = 100 \mu C$$

Q30: A car is moving on a plane inclined at 30° to the horizontal with an acceleration of 10 ms^{-2} parallel to the plane upward. A bob is suspended by a string from the roof of the car. The angle in degrees which the string makes with the vertical is _____.
(Take $g = 10 \text{ ms}^{-2}$)

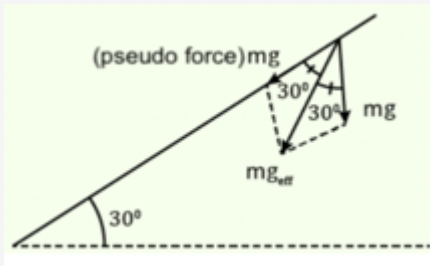
Q30: A car is moving on a plane inclined at 30° to the horizontal with an acceleration of 10 ms^{-2} parallel to the plane upward. A bob is suspended by a string from the roof of the car. The angle in degrees which the string makes with the vertical is _____.
(Take $g = 10 \text{ ms}^{-2}$)

30.00

Solution:

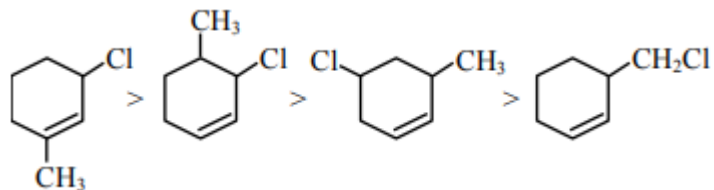
From frame of car, the effective acceleration (g_{eff}) due to gravity shall be measured as shown in figure.

Hence g_{eff} makes an angle 30° with vertical direction (down wards). Since the string aligns with direction of g_{eff} in equilibrium, the required angle is $\theta = 30^\circ$.

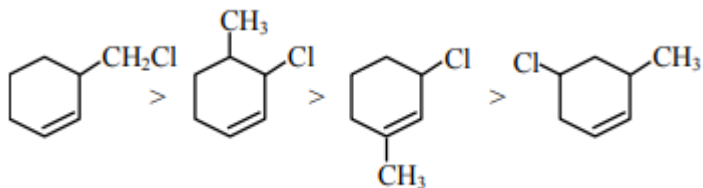


Q31: The correct order of reactivity of the given chlorides with acetate in acetic acid is

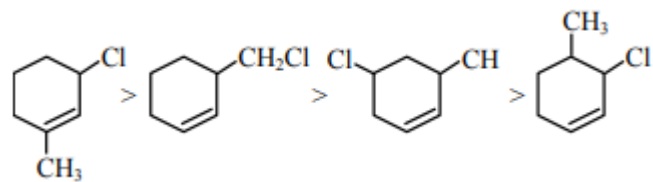
(A)



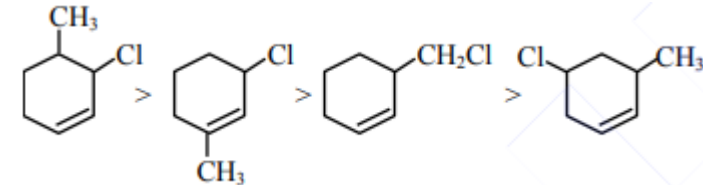
(B)



(C)

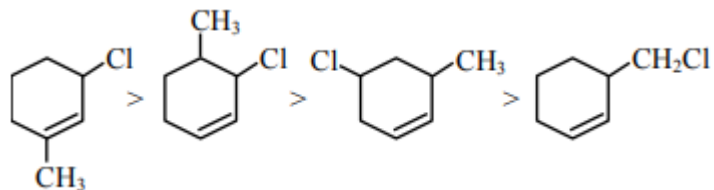


(D)

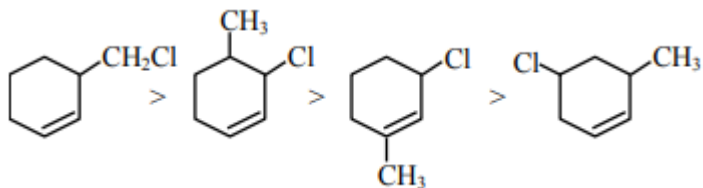


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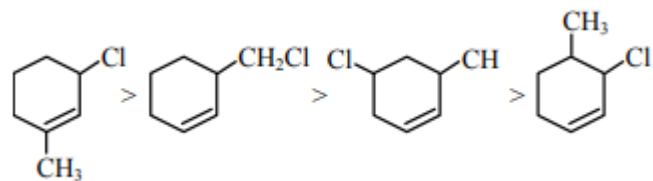
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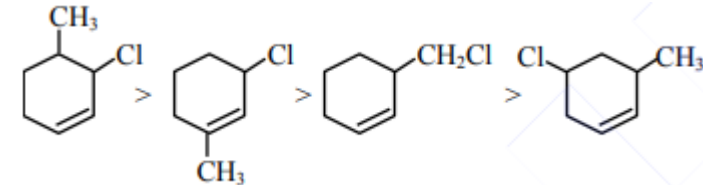
(B)



(C)

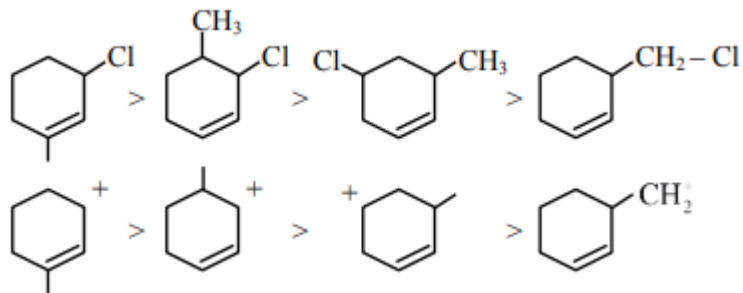


(D)

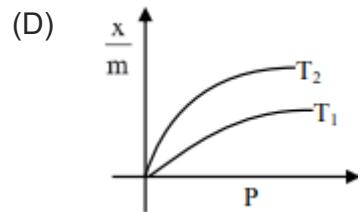
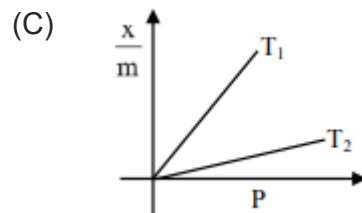
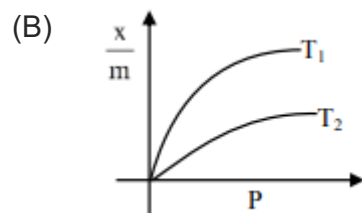
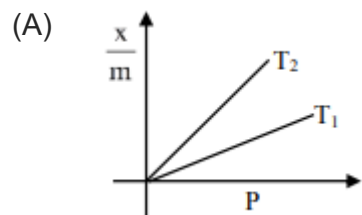


Solution:

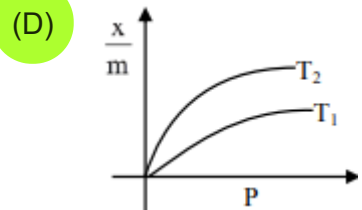
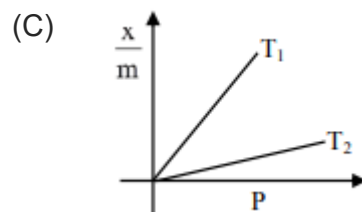
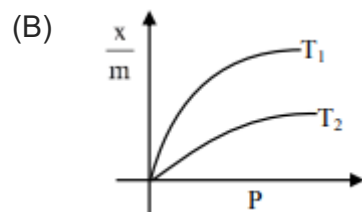
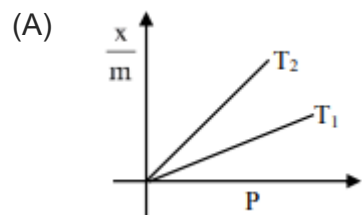
As it is example of S_N1 . So, carbocation stability $\uparrow$, reaction rate $\uparrow$.



Q32: Select the graph that correctly describes the adsorption isotherms at two temperatures T_1 and T_2 ($T_1 > T_2$) for a gas: (x – mass of the gas adsorbed; m – mass of adsorbent; P – pressure)



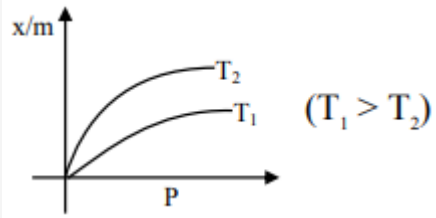
Q32: Select the graph that correctly describes the adsorption isotherms at two temperatures T_1 and T_2 ($T_1 > T_2$) for a gas: (x – mass of the gas adsorbed; m – mass of adsorbent; P – pressure)



Solution:

$$\frac{x}{m} \propto p^{\frac{1}{n}} \quad (0 < \frac{1}{n} < 1)$$

On Increasing temperature $\frac{x}{m}$ decreases since adsorption is generally exothermic.



Q33: The major component/ingredient of Portland Cement is :

- (A) tricalcium aluminate
- (B) tricalcium silicate
- (C) dicalcium aluminate
- (D) dicalcium silicate

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- (A) tricalcium aluminate
- ☒ (B) tricalcium silicate
- (C) dicalcium aluminate
- (D) dicalcium silicate

Solution:

Important ingredients present in Portland cement are dicalcium silicate (Ca_2SiO_4) 26%, tricalcium silicate (Ca_3SiO_5) 51% and tricalcium aluminate ($\text{Ca}_3\text{Al}_2\text{O}_6$) 11%.

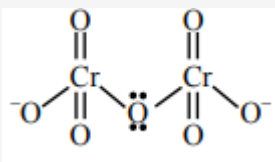
Q34: In the structure of the dichromate ion, there is a:

- (A) linear symmetrical Cr–O–Cr bond.
- (B) non-linear symmetrical Cr–O–Cr bond.
- (C) linear unsymmetrical Cr–O–Cr bond.
- (D) non-linear unsymmetrical Cr–O–Cr bond.

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- (A) linear symmetrical Cr–O–Cr bond.
- (B) non-linear symmetrical Cr–O–Cr bond.
- (C) linear unsymmetrical Cr–O–Cr bond.
- (D) non-linear unsymmetrical Cr–O–Cr bond.

Solution:



Dichromate ion contain non-linear symmetrical Cr–O–Cr bond.

Q35: Which one of the following compounds contains β -C₁ -C₄ glycosidic linkage ?

- (A) Lactose
- (B) Sucrose
- (C) Maltose
- (D) Amylose

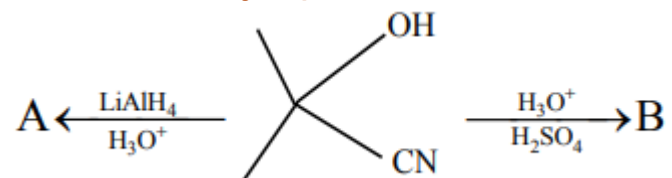
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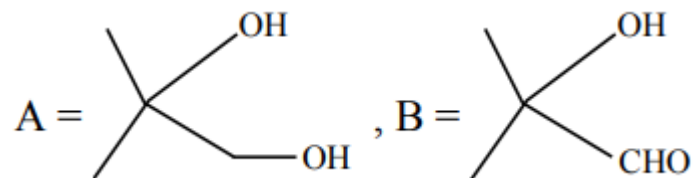
Solution:

In Lactose it is β C₁ – C₄ glycosidic linkage. In Maltose, Amylose α C₁ – C₄ glycosidic linkage is present.

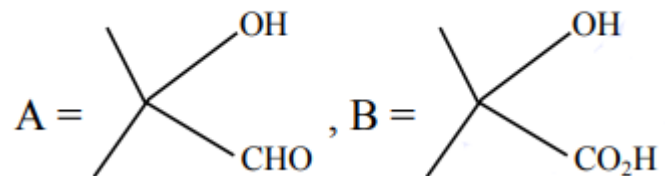
Q36: The major products A and B in the following set of reactions are



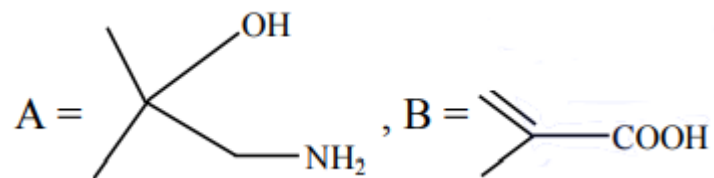
(A)



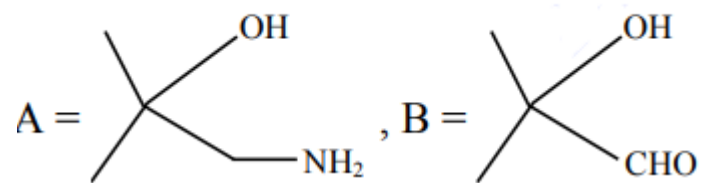
(B)



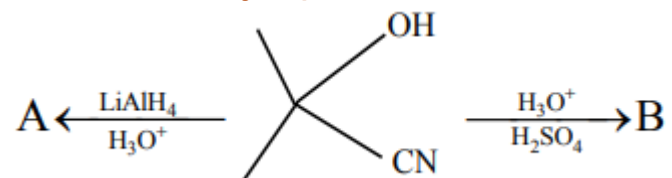
(C)



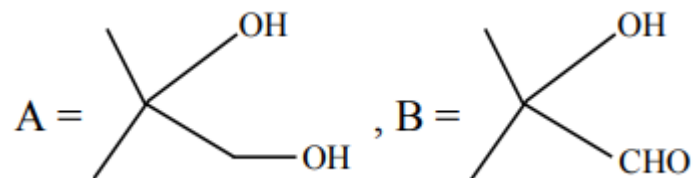
(D)



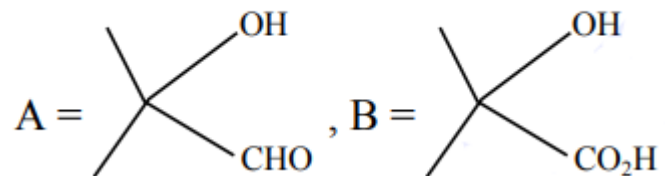
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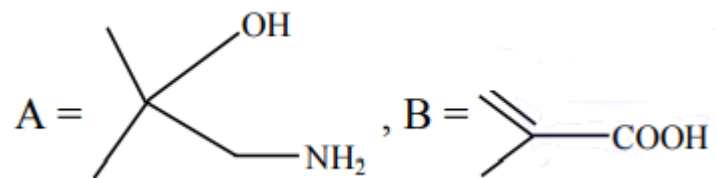
(A)



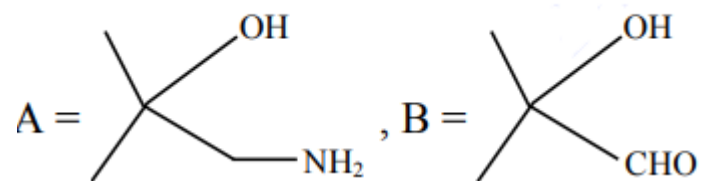
(B)



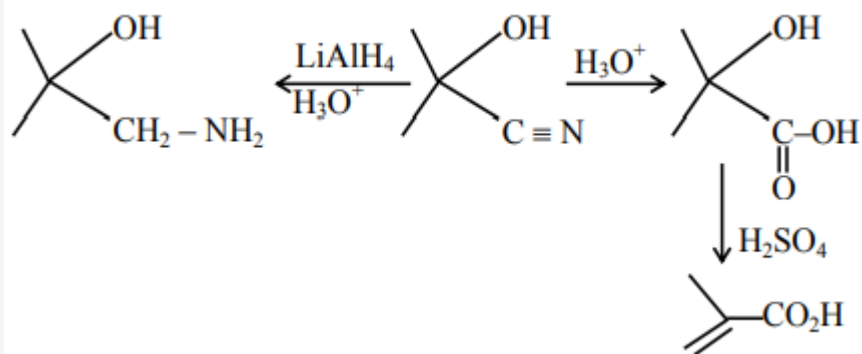
(C)



(D)



Solution:



Q37: Which one of the following lanthanides exhibits +2 oxidation state with diamagnetic nature?

(Given Z for Nd = 60, Yb = 70, La = 57, Ce = 58)

- (A) Nd
- (B) Yb
- (C) La
- (D) Ce

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(Given Z for Nd = 60, Yb = 70, La = 57, Ce = 58)

(A) Nd

(B) Yb

(C) La

(D) Ce

Solution:

Electronic configuration of Yb = $[\text{Xe}] 4f^{14} 6s^2$

Thus, electronic configuration of $\text{Yb}^{2+} = [\text{Xe}] 4f^{14}$

Since all the electrons are paired, the ion is diamagnetic.

Q38: Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Aluminium is extracted from bauxite by the electrolysis of molten mixture of Al_2O_3 with cryolite.

Reason (R): The oxidation state of Al in cryolite is +3.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) (A) is true but (R) is false
- (B) (A) is false but (R) is true
- (C) Both (A) and (R) are correct and (R) is the correct explanation of (A)
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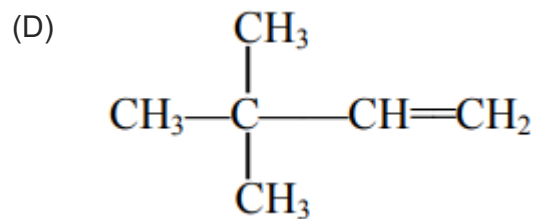
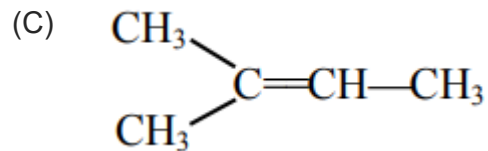
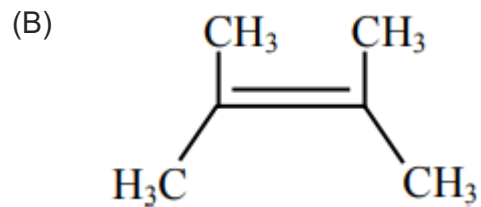
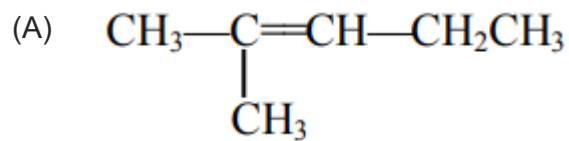
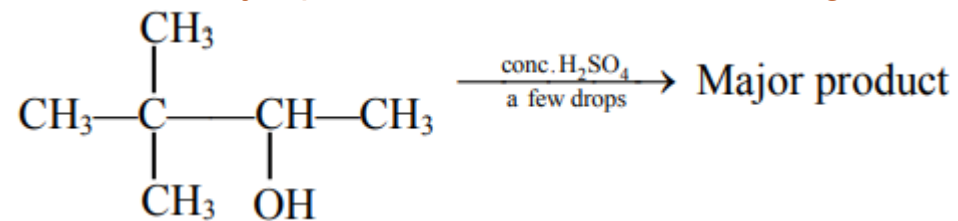
In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) (A) is true but (R) is false
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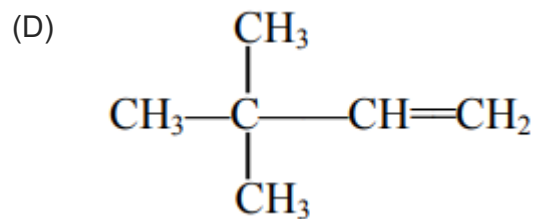
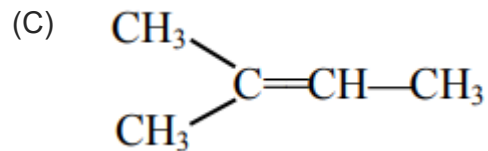
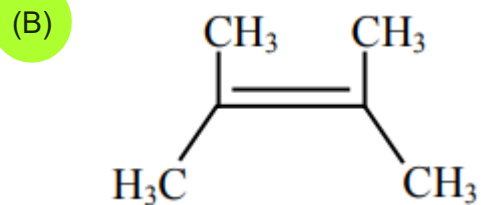
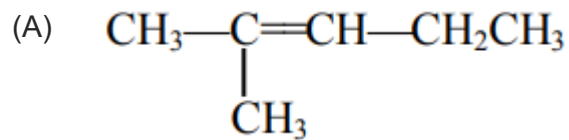
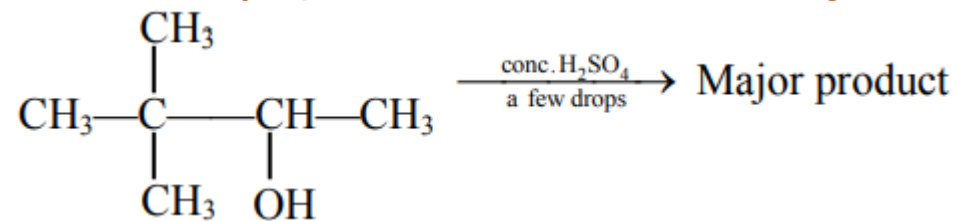
Solution:

- (A) Aluminium is reactive metal so Aluminium is extracted by electrolysis of Alumina with molten mixture of Cryolite.
- (B) Cryolite, Na_3AlF_6 - here Al is in +3 O.S.

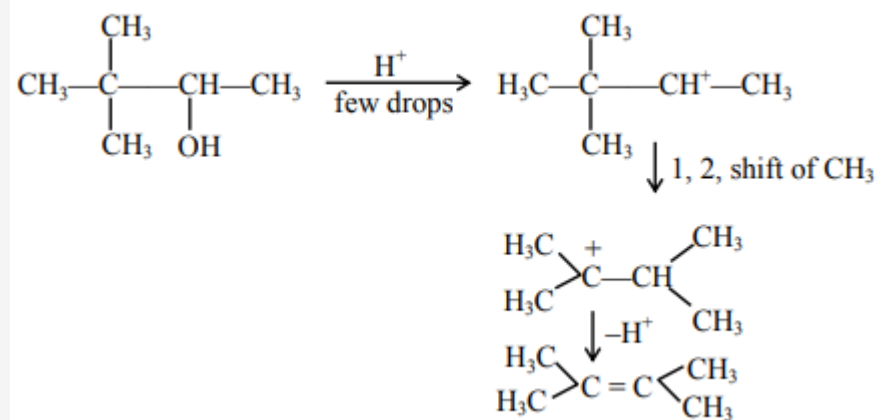
Q39: The major product formed in the following reaction is:



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Solution:



Q40: Monomer of Novolac is:

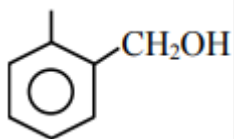
- (A) 3-Hydroxybutanoic acid
- (B) phenol and melamine
- (C) o-Hydroxymethylphenol
- (D) 1,3-Butadiene and styrene

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Solution:

Monomer of Novolac is



o-Hydroxymethylphenol

Q41: Given below are two statements:

Statement-I: The process of producing syn-gas is called gasification of coal.

Statement-II: The composition of syn-gas is $\text{CO} + \text{CO}_2 + \text{H}_2$ (1 : 1 : 1)

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) **Statement-I** is false, but **Statement-II** is true
- (B) **Statement-I** is true, but **Statement-II** is false
- (C) Both **Statement-I** and **Statement-II** are false
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- (C) Both **Statement-I** and **Statement-II** are false
- (D) Both **Statement-I** and **Statement-II** are true

Solution:

The process of producing syn-gas from coal is called gasification of coal.
Syn-gas having composition of CO & H_2 in 1:1

Q42: Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Treatment of bromine water with propene yields 1-bromopropan-2-ol.

Reason (R): Attack of water on bromonium ion follows Markovnikov rule and results in 1-bromopropan-2-ol.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Both **(A)** and **(R)** are true but **(R)** is NOT the correct explanation of **(A)**
- (B) **(A)** is false but **(R)** is true.
- (C) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**
- (D) **(A)** is true but **(R)** is false

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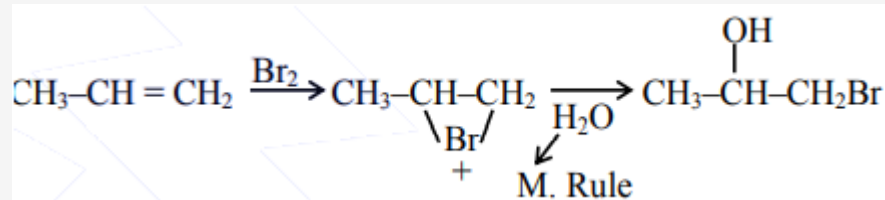
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- (B) **(A)** is false but **(R)** is true.
- ☒ (C) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**
- (D) **(A)** is true but **(R)** is false

Solution:



Its IUPAC name 1-bromopropan-2-ol.

A and R are true and (R) is the correct explanation of (A).

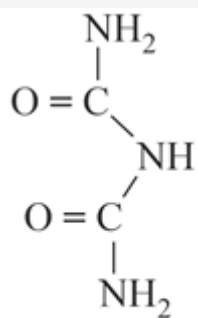
Q43: The denticity of an organic ligand, biuret is?

- (A) 2
- (B) 4
- (C) 3
- (D) 6

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- (A) 2
- (B) 4
- (C) 3
- (D) 6

Solution:



Biuret

It is a bidentate ligand as it can bind through both terminal N atoms at the same time.

Q44: Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Metallic character decreases and non-metallic character increases on moving from left to right in a period.

Reason (R): It is due to increase in ionisation enthalpy and decrease in electron gain enthalpy when one moves from left to right in a period.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) **(A)** is false but **(R)** is true.
- (B) **(A)** is true but **(R)** is false
- (C) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
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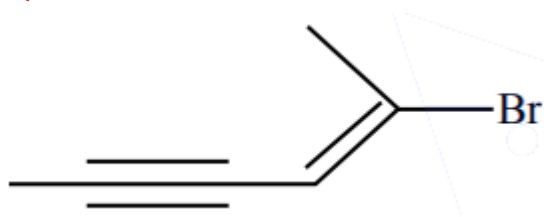
In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) (A) is false but (R) is true.
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Solution:

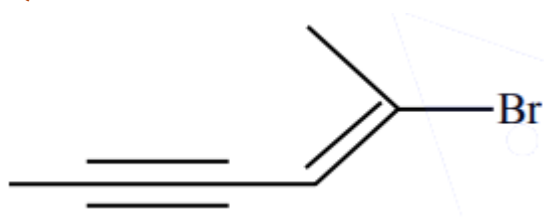
From left to right in periodic table - Metallic character decreases, Non-metallic character increases
⇒ It is due to increase in ionization enthalpy and increase in electron gain enthalpy

Q45: Choose the correct name for compound given below :



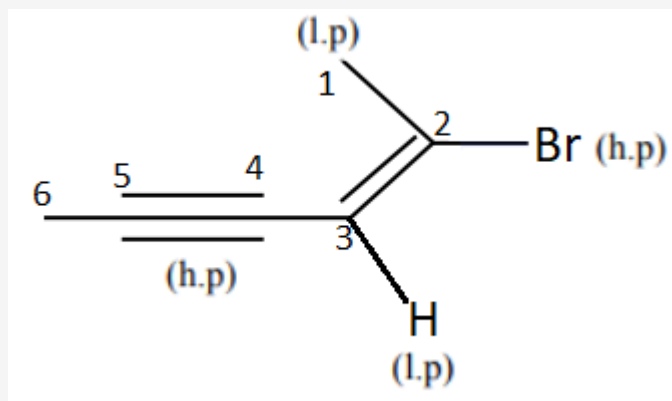
- (A) (4E)-5-Bromo-hex-4-en-2-yne
- (B) (2E)-2-Bromo-hex-4-yn-2-ene
- (C) (2E)-2-Bromo-hex-2-en-4-yne
- (D) (4E)-5-Bromo-hex-2-en-4-yne

Q45: Choose the correct name for compound given below :



- (A) (4E)-5-Bromo-hex-4-en-2-yne
- (B) (2E)-2-Bromo-hex-4-yn-2-ene
- (C) (2E)-2-Bromo-hex-2-en-4-yne
- (D) (4E)-5-Bromo-hex-2-en-4-yne

Solution:



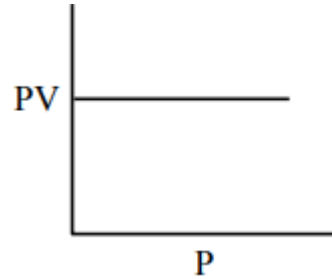
h.p. $\Rightarrow$ higher priority

l.p. $\Rightarrow$ lower priority

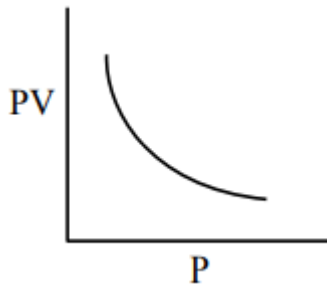
(2E)-2-bromo-2-hexene

Q46: Which one of the following is the correct PV vs P plot at constant temperature for an ideal gas? (P and V stand for pressure and volume of the gas respectively).

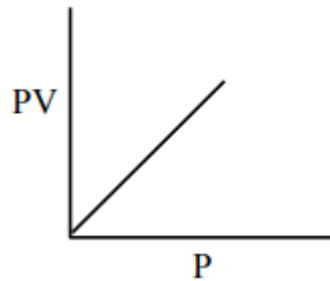
(A)



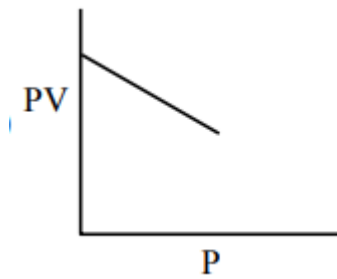
(B)



(C)

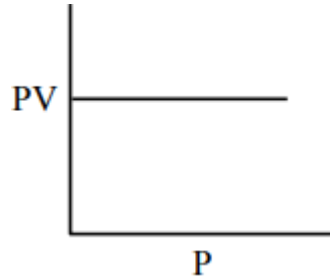


(D)

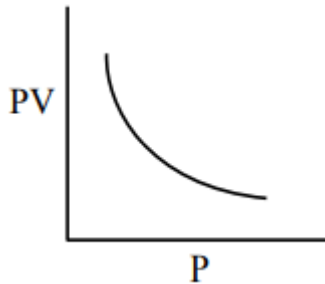


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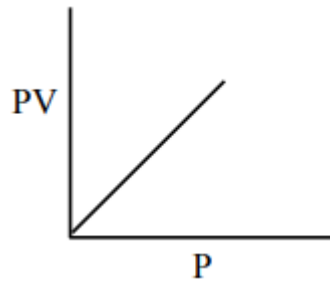
(A)



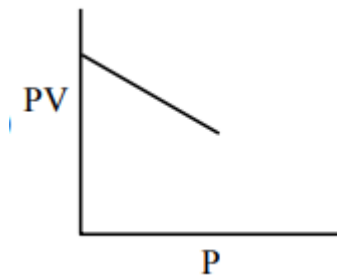
(B)



(C)



(D)

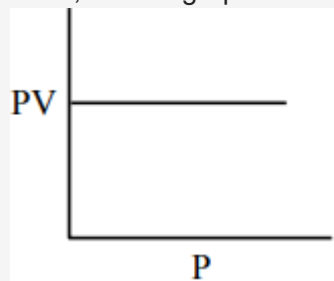


Solution:

$$PV = nRT \text{ (n, T constant)}$$

$$PV = \text{constant}$$

Thus, correct graph is:



Q47: Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): A simple distillation can be used to separate a mixture of propanol and propanone.

Reason (R): Two liquids with a difference of more than 20°C in their boiling points can be separated by simple distillations.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) **(A)** is false but **(R)** is true.
- (B) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.
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Solution:

A method of separating mixtures based on differences in their volatilities in a boiling liquid mixture. Two liquids with a difference of more than 20°C in their boiling points can be separated by simple distillations.

Both assertion & reason are correct & (R) is the correct explanation of (A)

Q48: Which one of the following 0.10 M aqueous solutions will exhibit the largest freezing point depression?

- (A) hydrazine
- (B) glucose
- (C) glycine
- (D) KHSO_4

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- (A) hydrazine
- (B) glucose
- (C) glycine
- (D) KHSO_4

Solution:

Since van't Hoff factor is highest for KHSO_4 , therefore colligative property (ΔT_f) will be highest for KHSO_4 .

Q49: BOD values (in ppm) for clean water (A) and polluted water (B) are expected respectively :

(A) $A > 50, B < 27$

(B) $A > 25, B < 17$

(C) $A < 5, B > 17$

(D) $A > 15, B > 47$

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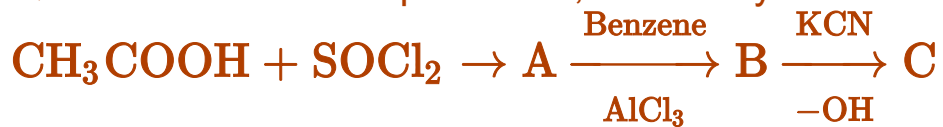
(D) $A > 15, B > 47$

Solution:

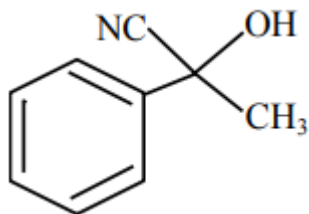
BOD values of clean water (A) is less than 5 ppm. So, $A < 5$

BOD values of polluted water (B) is greater than 17 ppm. So $B > 17$

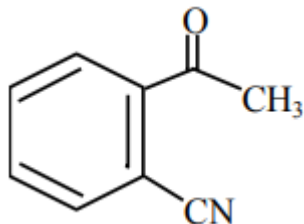
Q50: The structure of product C, formed by the following sequence of reactions is.



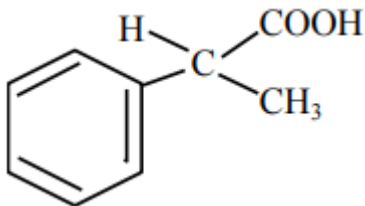
(A)



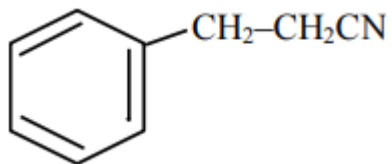
(B)



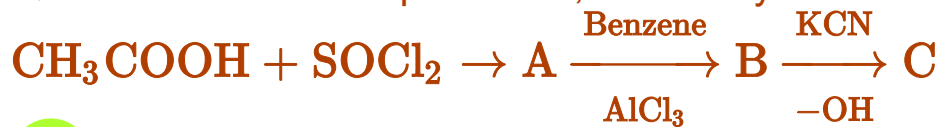
(C)



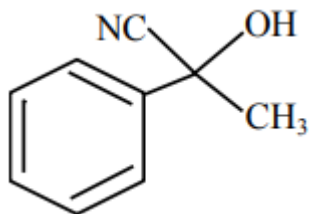
(D)



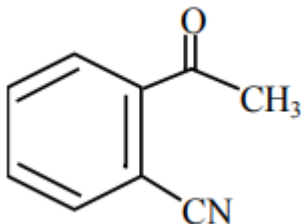
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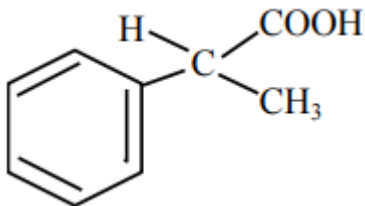
(A)



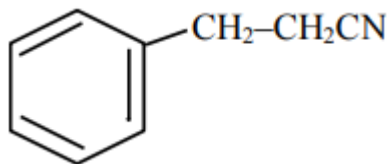
(B)



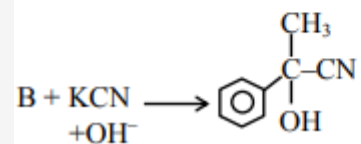
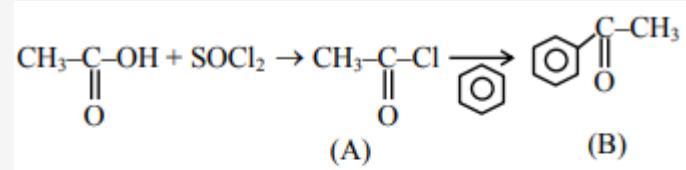
(C)



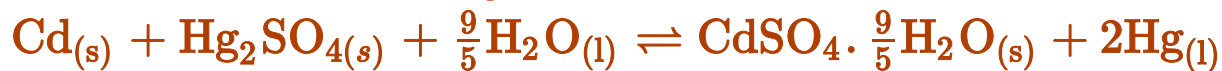
(D)



Solution:



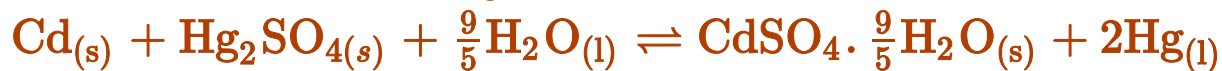
Q51: Consider the following cell reaction :



The value of E_{Cell}^0 is 4.315 V at 25°C. If $\Delta H^\circ = -825.2 \text{ kJ mol}^{-1}$, the standard entropy change ΔS° in J K^{-1} is _____. (Nearest integer)

[Given : Faraday constant = 96487 C mol^{-1}]

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The value of E_{Cell}^0 is 4.315 V at 25°C. If $\Delta H^\circ = -825.2 \text{ kJ mol}^{-1}$, the standard entropy change ΔS° in J K^{-1} is _____. (Nearest integer)

[Given : Faraday constant = 96487 C mol^{-1}]

25.00

Solution:

$$\Delta G^\circ = -nFE^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta S^\circ = \frac{\Delta H^\circ + nFE^\circ}{T}$$

$$\Delta S^\circ = \frac{(-825.2 \times 10^3) + (2 \times 96487 \times 4.315)}{298}$$

$$\Delta S^\circ = \frac{-825.2 \times 10^3 + 832.682 \times 10^3}{298}$$

$$\Delta S^\circ = \frac{7.483 \times 10^3}{298} = 25.11 \text{ JK}^{-1}\text{mol}^{-1}$$

Q52: The molarity of the solution prepared by dissolving 6.3 g of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) in 250 mL of water in mol L^{-1} is $x \times 10^{-2}$. The value of x is _____. (Nearest integer)

[Atomic mass H : 1.0, C : 12.0, O : 16.0]

Q52: The molarity of the solution prepared by dissolving 6.3 g of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) in 250 mL of water in mol L^{-1} is $x \times 10^{-2}$. The value of x is _____. (Nearest integer)
[Atomic mass H : 1.0, C : 12.0, O : 16.0]

20.00

Solution:

$$[\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}] = \frac{\text{weight/MW}}{V(\text{L})}$$

$$\Rightarrow x \times 10^{-2} = \frac{6.3/126}{250/1000}$$

$$x = 20$$

Q53: Consider the sulphides - HgS, PbS, CuS, Sb₂S₃, As₂S₃ and CdS. Number of these sulphides soluble in 50% HNO₃ is _____.

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4.00

Solution:

Pbs, CuS, As₂S₃ , CdS are soluble in 50% HNO₃. HgS, Sb₂S₃ are insoluble in 50% HNO₃.

Q54: The total number of reagents from those given below, that can convert nitrobenzene into aniline is _____. (Integer answer)

I. Sn – HCl

II. Sn – NH_4OH

III. Fe – HCl

IV. Zn – HCl

V. H_2 – Pd

VI. H_2 – Raney Nickel

Q54: The total number of reagents from those given below, that can convert nitrobenzene into aniline is _____. (Integer answer)

I. Sn – HCl

II. Sn – NH_4OH

III. Fe – HCl

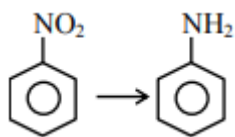
IV. Zn – HCl

V. H_2 – Pd

VI. H_2 – Raney Nickel

5.00

Solution:



Reagents used can be

(i) Sn + HCl

(ii) Fe + HCl

(iii) Zn + HCl

(iv) H_2 – Pd

(v) H_2 (Raney Ni)

Q55: The number of halogen(s) forming halic (V) acid is _____.

Q55: The number of halogen(s) forming halic (V) acid is _____.

3.00

Solution:

The number of halogens forming halic (V) acid is three - HClO_3 HBrO_3 HIO_3 .

Q56: For a first order reaction, the ratio of the time for 75% completion of a reaction to the time for 50% completion is _____. (Integer answer)

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Solution:

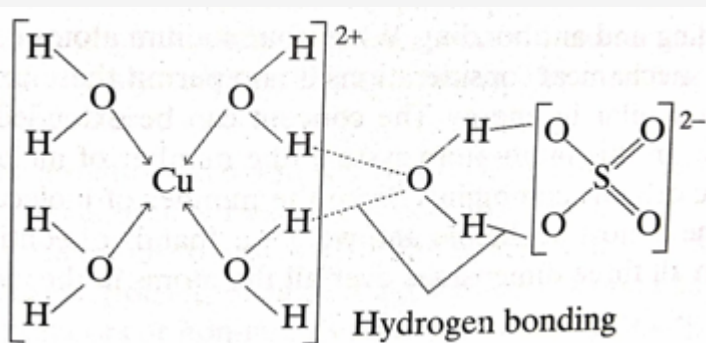
After 75% completion of a reaction, 25% of the reactant remains. Thus, 75% completion corresponds to two half-lives.

Q57: The number of hydrogen bonded water molecule(s) associated with stoichiometry $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is _____.

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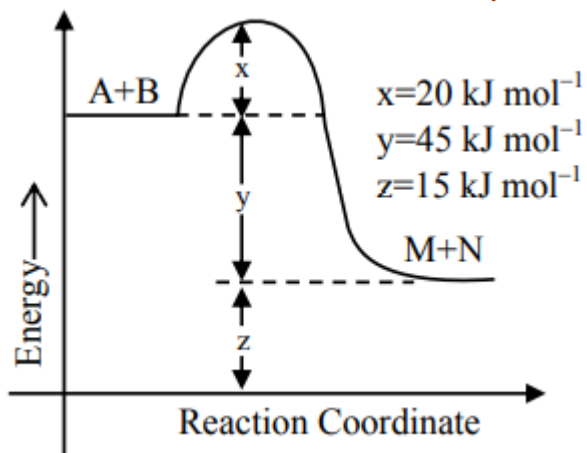
1.00

Solution:

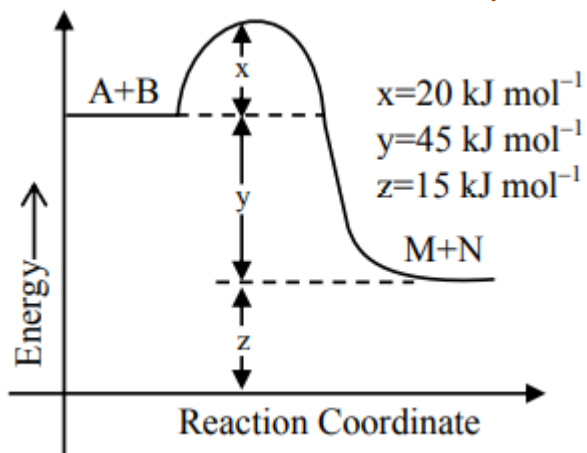


One hydrogen bonded H_2O molecule.

Q58: According to the following figure, the magnitude of the enthalpy change of the reaction $A + B \rightarrow M + N$ in kJ mol^{-1} is equal to _____. (Integer answer)



Q58: According to the following figure, the magnitude of the enthalpy change of the reaction $A + B \rightarrow M + N$ in kJ mol^{-1} is equal to _____. (Integer answer)



45.00

Solution:

$$\Delta H = E_{\text{products}} - E_{\text{reactants}} = -y = -45 \text{ kJ mol}^{-1}$$

Q59: Ge ($Z = 32$) in its ground state electronic configuration has x completely filled orbitals with $m_l = 0$. The value of x is _____.

Q59: Ge ($Z = 32$) in its ground state electronic configuration has x completely filled orbitals with $m_l = 0$. The value of x is _____.

7.00

Solution:

The electronic configuration is:

1s²	2s²	2p⁶			3s²	3p⁶			4s²	3d¹⁰					4p²		
1↓	1↓	1↓	1↓	1↓	1↓	1↓	1↓	1↓	1↓	1↓	1↓	1↓	1↓	1↓	1	1	
0	0	-1	0	+1	0	-1	0	+1	0	-2	-1	0	+1	+2	-1	0	+1

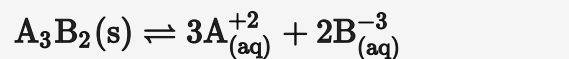
Thus, number of completely filled orbitals with $m_l = 0$ is 7.

Q60: A_3B_2 is a sparingly soluble salt of molar mass M (g mol^{-1}) and solubility x g L^{-1} . The solubility product satisfies $K_{\text{sp}} = a\left(\frac{x}{M}\right)^5$. The value of a is _____. (Integer Answer)

Q60: A_3B_2 is a sparingly soluble salt of molar mass M (g mol^{-1}) and solubility x g L^{-1} . The solubility product satisfies $K_{\text{sp}} = a \left(\frac{x}{M} \right)^5$. The value of a is _____. (Integer Answer)

108.00

Solution:



$3s \quad 2s$

$$K_{\text{SP}} = (3s)^3 (2s)^2$$

$$K_{\text{SP}} = 108 s^5 \text{ \& } s = (x/M)$$

$$K_{\text{SP}} = 108 \left(\frac{x}{m} \right)^5$$

$$\text{given } K_{\text{sp}} = a \left(\frac{x}{m} \right)^5$$

$$\text{Comparing } a = 108$$

Q61: Which of the following is not correct for relation R on the set of real numbers?

- (A) $(x, y) \in R \Leftrightarrow |x - y| \leq 1$ is reflexive and symmetric.
- (B) $(x, y) \in R \Leftrightarrow 0 < |x - y| \leq 1$ is symmetric but not transitive.
- (C) $(x, y) \in R \Leftrightarrow |x| - |y| \leq 1$ is reflexive but not symmetric.
- (D) $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$ is neither transitive nor symmetric.

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- (D) $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$ is neither transitive nor symmetric.

Solution:

Note that $(1, 2)$ and $(2, 3)$ satisfy $0 < |x - y| \leq 1$
 $0 \leq |x - y| \leq 1$ is symmetric but not transitive.

Q62: Three numbers are in an increasing geometric progression with common ratio r . If the middle number is doubled, then the new numbers are in arithmetic progression with common difference d . If the fourth term of GP is $3r^2$, then $r^2 - d$ is equal to?

(A) $7 + 3\sqrt{3}$

(B) $7 - \sqrt{3}$

(C) $7 - 7\sqrt{3}$

(D) $7 + \sqrt{3}$

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(A) $7 + 3\sqrt{3}$

(B) $7 - \sqrt{3}$

(C) $7 - 7\sqrt{3}$

(D) $7 + \sqrt{3}$

Solution:

Let number be $\frac{a}{r}, a, ar \rightarrow G.P$

$$\frac{a}{r}, 2a, ar \rightarrow A.P \Rightarrow 4a = \frac{a}{r} + ar \Rightarrow r + \frac{1}{r} = 4$$

$$r = 2 \pm \sqrt{3}$$

$$4^{\text{th}} \text{ term of G.P.} = 3r^2 \Rightarrow ar^2 = 3r^2 \Rightarrow a = 3$$

$$r = 2 + \sqrt{3}, a = 3, d = 2a - \frac{a}{r} = 3\sqrt{3}$$

$$r^2 - d = (2 + \sqrt{3})^2 - 3\sqrt{3}$$

$$= 7 + 4\sqrt{3} - 3\sqrt{3}$$

$$= 7 + \sqrt{3}.$$

Q63: The sum of 10 terms of the series $\frac{3}{1^2 \times 2^2} + \frac{5}{2^2 \times 3^2} + \frac{7}{3^2 \times 4^2} + \dots$ is?

(A) $\frac{120}{121}$

(B) 1

(C) $\frac{143}{144}$

(D) $\frac{99}{100}$

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(A) $\frac{120}{121}$

(B) 1

(C) $\frac{143}{144}$

(D) $\frac{99}{100}$

Solution:

$$\begin{aligned} S &= \frac{2^2-1^2}{1^2 \times 2^2} + \frac{3^2-2^2}{2^2 \times 3^2} + \frac{4^2-3^2}{3^2 \times 4^2} + \dots \\ &= \left[\frac{1}{1^2} - \frac{1}{2^2} \right] + \left[\frac{1}{2^2} - \frac{1}{3^2} \right] + \left[\frac{1}{3^2} - \frac{1}{4^2} \right] + \dots + \left[\frac{1}{10^2} - \frac{1}{11^2} \right] \\ &= 1 - \frac{1}{121} \\ &= \frac{120}{121} \end{aligned}$$

Q64: The number of real roots of the equation $e^{4x} + 2e^{3x} - e^x - 6 = 0$ is?

(A) 0

(B) 1

(C) 4

(D) 2

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(A) 0

(B) 1

(C) 4

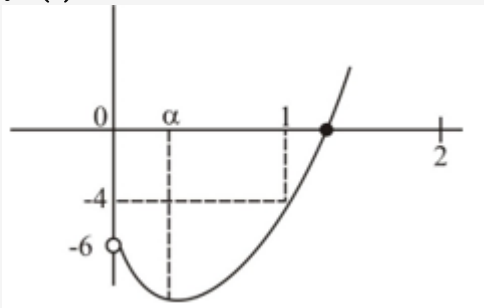
(D) 2

Solution:

Let $e^x = t > 0$

$$f(t) = t^4 + 2t^3 - t - 6 = 0$$

$$f'(t) = 4t^3 + 6t^2 - 1$$



$$f''(t) = 12t^2 + 12t > 0$$

$$f(0) = -6, f(1) = -4, f(2) = 24$$

$\Rightarrow$ Numbers of real root = 1.

Q65: The integral $\int \frac{1}{\sqrt[4]{(x-1)^3(x+2)^5}} dx$ is equal to

(where C is a constant of integration)

(A) $\frac{3}{4} \left(\frac{x+2}{x-1} \right)^{\frac{5}{4}} + c$

(B) $\frac{3}{4} \left(\frac{x+2}{x-1} \right)^{\frac{1}{4}} + c$

(C) $\frac{4}{3} \left(\frac{x-1}{x+2} \right)^{\frac{1}{4}} + c$

(D) $\frac{4}{3} \left(\frac{x-1}{x+2} \right)^{\frac{5}{4}} + c$

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(C) $\frac{4}{3} \left(\frac{x-1}{x+2} \right)^{\frac{1}{4}} + c$

(D) $\frac{4}{3} \left(\frac{x-1}{x+2} \right)^{\frac{5}{4}} + c$

Solution:

$$\int \frac{dx}{(x-1)^{3/4}(x+2)^{5/4}}$$
$$= \int \frac{dx}{\left(\frac{x+2}{x-1}\right)^{5/4} \cdot (x-1)^2}$$

put $\frac{x+2}{x-1} = t$

$$= -\frac{1}{3} \int \frac{dt}{t^{5/4}}$$

$$= \frac{4}{3} \cdot \frac{1}{t^{1/4}} + c$$

$$= \frac{4}{3} \left(\frac{x-1}{x+2} \right)^{1/4} + c.$$

Q66: A vertical pole fixed to the horizontal ground is divided in the ratio **3 : 7** by a mark on it with lower part shorter than the upper part. If the two parts subtend equal angles at a point on the ground 18m away from the base of the pole, then the height of the pole (in meters) is?

(A) $12\sqrt{15}$

(B) $8\sqrt{10}$

(C) $6\sqrt{10}$

(D) $12\sqrt{10}$

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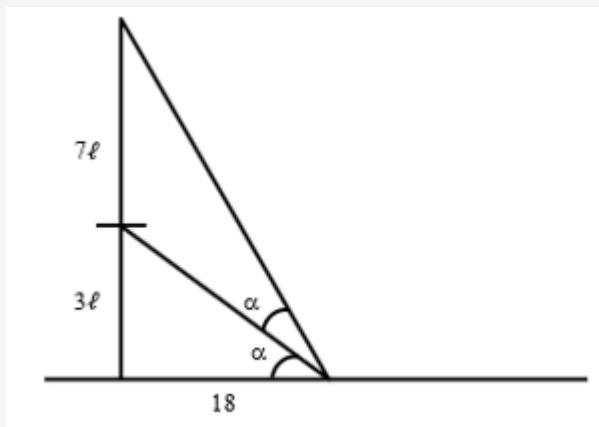
(A) $12\sqrt{15}$

(B) $8\sqrt{10}$

(C) $6\sqrt{10}$

(D) $12\sqrt{10}$

Solution:



Let height of pole = $10l$

$$\tan \alpha = \frac{3l}{18} = \frac{l}{6}$$

$$\tan 2\alpha = \frac{10l}{18}$$

$$\frac{2 \tan \alpha}{1 - \tan^2 \alpha} = \frac{10l}{18}$$

$$\text{Use } \tan \alpha = \frac{l}{6} \Rightarrow l = \sqrt{\frac{72}{5}}$$

$$\text{Height of pole} = 10l = 12\sqrt{10}$$

Q67: Let f be a non-negative function in $[0, 1]$ and twice differentiable in $(0, 1)$. If

$$\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt, 0 \leq x \leq 1 \text{ and } f(0) = 0, \text{ then } \lim_{x \rightarrow 0} \frac{1}{x^2} \int_0^x f(t) dt :$$

- (A) Equals 1
- (B) Does not exist
- (C) Equals $\frac{1}{2}$
- (D) Equals 0

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- (A) Equals 1
- (B) Does not exist
- (C) Equals $\frac{1}{2}$
- (D) Equals 0

Solution:

$$\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt, 0 \leq x \leq 1$$

Differentiating both the sides

$$\sqrt{1 - (f'(x))^2} = f(x)$$

$$\Rightarrow 1 - (f'(x))^2 = f^2(x)$$

$$\frac{f'(x)}{\sqrt{1 - f^2(x)}} = 1$$

$$\sin^{-1} f(x) = x + c$$

$$\because f(0) = 0 \Rightarrow c = 0 \Rightarrow f(x) = \sin x$$

$$\text{Now } \lim_{x \rightarrow 0} \frac{\int_0^x \sin t dt}{x^2} = \frac{1}{2}$$

Q68: If the function $f(x) = \begin{cases} \frac{1}{x} \log_e \left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}} \right), & x < 0 \\ k, & x = 0 \\ \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2+1}-1}, & x > 0 \end{cases}$ is continuous at $x = 0$, then

$\frac{1}{a} + \frac{1}{b} + \frac{4}{k}$ is equal to ?

(A) -4

(B) -5

(C) 4

(D) 5

Q68: If the function $f(x) = \begin{cases} \frac{1}{x} \log_e \left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}} \right), & x < 0 \\ k, & x = 0 \\ \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2+1}-1}, & x > 0 \end{cases}$ is continuous at $x = 0$, then

$\frac{1}{a} + \frac{1}{b} + \frac{4}{k}$ is equal to ?

(A) -4

(B) -5

(C) 4

(D) 5

Solution:

If $f(x)$ is continuous at $x = 0$, $RHL = LHL = f(0)$

$$\begin{aligned}\lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2 + 1} - 1} \cdot \frac{\sqrt{x^2 + 1} + 1}{\sqrt{x^2 + 1} + 1} \quad (\text{Rationalisation}) \\ &= \lim_{x \rightarrow 0^+} -\frac{2\sin^2 x}{x^2} \cdot (\sqrt{x^2 + 1} + 1) = -4\end{aligned}$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{1}{x} \ln \left(\frac{1 + \frac{x}{a}}{1 - \frac{x}{b}} \right)$$

$$\lim_{x \rightarrow 0^-} \frac{\ln(1 + \frac{x}{a})}{(\frac{x}{a}) \cdot a} + \frac{\ln(1 - \frac{x}{b})}{(-\frac{x}{b}) \cdot b}$$

$$= \frac{1}{a} + \frac{1}{b}$$

$$\text{So } \frac{1}{a} + \frac{1}{b} = -4 = k$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{4}{k} = -4 - 1 = -5.$$

Q69: The function $f(x) = |x^2 - 2x - 3| \cdot e^{9x^2 - 12x + 4}$ is not differentiable at exactly ?

- (A) Two points
- (B) One points
- (C) Four points
- (D) Three points

Q69: The function $f(x) = |x^2 - 2x - 3| \cdot e^{9x^2 - 12x + 4}$ is not differentiable at exactly ?

- (A) Two points
- (B) One points
- (C) Four points
- (D) Three points

Solution:

$$f(x) = |(x-3)(x+1)| \cdot e^{(3x-2)^2}$$
$$f(x) = \begin{cases} (x-3)(x+1) \cdot e^{(3x-2)^2} & ; \quad x \in (3, \infty) \\ -(x-3)(x+1) \cdot e^{(3x-2)^2} & ; \quad x \in [-1, 3] \\ (x-3) \cdot (x+1) \cdot e^{(3x-2)^2} & ; \quad x \in (-\infty, -1) \end{cases}$$

Clearly, not-differentiable at $x = -1$ & $x = 3$.

Q70: If $a_r = \cos \frac{2r\pi}{9} + i \sin \frac{2r\pi}{9}$, $r = 1, 2, 3, \dots, i = \sqrt{-1}$, then the determinant

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix} \text{ is equal to?}$$

(A) $a_2 a_6 - a_4 a_8$

(B) $a_1 a_9 - a_3 a_7$

(C) a_5

(D) a_9

Q70: If $a_r = \cos \frac{2r\pi}{9} + i \sin \frac{2r\pi}{9}$, $r = 1, 2, 3, \dots$, $i = \sqrt{-1}$, then the determinant

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix} \text{ is equal to?}$$

(A) $a_2 a_6 - a_4 a_8$

(B) $a_1 a_9 - a_3 a_7$

(C) a_5

(D) a_9

Solution:

$a_r = e^{\frac{i2\pi r}{9}}$, $r = 1, 2, 3, \dots$ $a_1, a_2, a_3, \dots$ are in G.P.

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix} = \begin{vmatrix} a_1 & a_1^2 & a_1^3 \\ a_1^4 & a_1^5 & a_1^6 \\ a_1^7 & a_1^8 & a_1^9 \end{vmatrix} = a_1 \cdot a_1^4 \cdot a_1^7 \begin{vmatrix} 1 & a_1 & a_1^2 \\ 1 & a_1 & a_1^2 \\ 1 & a_1 & a_1^2 \end{vmatrix} = 0$$

Now $a_1 a_9 - a_3 a_7 = a_1^{10} - a_1^{10} = 0$

Q71: Let the equation of the plane, that passes through the point $(1, 4, -3)$ and contains the line of intersection of the planes $3x - 2y + 4z - 7 = 0$ and $x + 5y - 2z + 9 = 0$, be $\alpha x + \beta y + \gamma z + 3 = 0$ then $\alpha + \beta + \gamma$ is equal to ?

- (A) -15
- (B) 15
- (C) -23
- (D) 23

Q71: Let the equation of the plane, that passes through the point $(1, 4, -3)$ and contains the line of intersection of the planes $3x - 2y + 4z - 7 = 0$ and $x + 5y - 2z + 9 = 0$, be $\alpha x + \beta y + \gamma z + 3 = 0$ then $\alpha + \beta + \gamma$ is equal to ?

(A) -15

(B) 15

(C) -23

(D) 23

Solution:

Equation of plane is

$$3x - 2y + 4z - 7 + \lambda(x + 5y - 2z + 9) = 0$$

$$(3 + \lambda)x + (5\lambda - 2)y + (4 - 2\lambda)z + 9\lambda - 7 = 0$$

passing through $(1, 4, -3)$

$$\Rightarrow 3 + \lambda + 20\lambda - 8 - 12 + 6\lambda + 9\lambda - 7 = 0$$

$$\Rightarrow \lambda = \frac{2}{3}$$

$\Rightarrow$ equation of plane is

$$-11x - 4y - 8z + 3 = 0$$

$$\Rightarrow \alpha + \beta + \gamma = -23.$$

Q72: The line $12x \cos \theta + 5y \sin \theta = 60$ is tangent to which of the following curves ?

(A) $25x^2 + 12y^2 = 3600$

(B) $144x^2 + 25y^2 = 3600$

(C) $x^2 + y^2 = 169$

(D) $x^2 + y^2 = 60$

Q72: The line $12x \cos \theta + 5y \sin \theta = 60$ is tangent to which of the following curves ?

(A) $25x^2 + 12y^2 = 3600$

(B) $144x^2 + 25y^2 = 3600$

(C) $x^2 + y^2 = 169$

(D) $x^2 + y^2 = 60$

Solution:

$$12x \cos \theta + 5y \sin \theta = 60$$

$$\frac{x \cos \theta}{5} + \frac{y \sin \theta}{12} = 1$$

is tangent to $\frac{x^2}{25} + \frac{y^2}{144} = 1$

$$144x^2 + 25y^2 = 3600.$$

Q73: If p and q are the length of the perpendiculars from the origin on the lines $x \operatorname{cosec} \alpha - y \sec \alpha = k \cot 2\alpha$ and $x \sin \alpha + y \cos \alpha = k \sin 2\alpha$ respectively, then k^2 is equal to ?

(A) $p^2 + 2q^2$

(B) $2p^2 + q^2$

(C) $p^2 + 4q^2$

(D) $4p^2 + q^2$

Q73: If p and q are the length of the perpendiculars from the origin on the lines $x \operatorname{cosec} \alpha - y \sec \alpha = k \cot 2\alpha$ and $x \sin \alpha + y \cos \alpha = k \sin 2\alpha$ respectively, then k^2 is equal to ?

(A) $p^2 + 2q^2$

(B) $2p^2 + q^2$

(C) $p^2 + 4q^2$

(D) $4p^2 + q^2$

Solution:

First line is $\frac{x}{\sin \alpha} - \frac{y}{\cos \alpha} = \frac{k \cos 2\alpha}{\sin 2\alpha}$

$$\Rightarrow x \cos \alpha - y \sin \alpha = \frac{k}{2} \cos 2\alpha$$

$$\Rightarrow p = \left| \frac{k}{2} \cos 2\alpha \right| \Rightarrow 2p = |k \cos 2\alpha| \quad \dots (i)$$

Second line is $x \sin \alpha + y \cos \alpha = k \sin 2\alpha$

$$\Rightarrow q = |k \sin 2\alpha| \quad \dots (ii)$$

Hence $4p^2 + q^2 = k^2$ (from (i) and (ii)).

Q74: $\lim_{x \rightarrow 0} \frac{\sin^2(\pi \cos^4 x)}{x^4}$ is equal to ?

(A) $2\pi^2$

(B) 4π

(C) π^2

(D) $4\pi^2$

Q74: $\lim_{x \rightarrow 0} \frac{\sin^2(\pi \cos^4 x)}{x^4}$ is equal to ?

(A) $2\pi^2$

(B) 4π

(C) π^2

(D) $4\pi^2$

Solution:

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\sin^2(\pi \cos^4 x)}{x^4} \\ &= \lim_{x \rightarrow 0} \frac{1 - \cos(2\pi \cos^4 x)}{2x^4} \\ &= \lim_{x \rightarrow 0} \frac{1 - \cos(2\pi - 2\pi \cos^4 x)}{[2\pi(1 - \cos^4 x)]^2} 4\pi^2 \cdot \frac{\sin^4 x}{2x^4} (1 + \cos^2 x)^2 \\ &= \frac{1}{2} \cdot 4\pi^2 \cdot \frac{1}{2} (2)^2 = 4\pi^2 \end{aligned}$$

Q75: $\operatorname{cosec} 18^\circ$ is a root of the equation :

(A) $x^2 - 2x - 4 = 0$

(B) $x^2 - 2x + 4 = 0$

(C) $4x^2 + 2x - 1 = 0$

(D) $x^2 + 2x - 4 = 0$

Q75: cosec 18° is a root of the equation :

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(B) $x^2 - 2x + 4 = 0$

(C) $4x^2 + 2x - 1 = 0$

(D) $x^2 + 2x - 4 = 0$

Solution:

$$\text{cosec } 18^\circ = \frac{1}{\sin 18^\circ} = \frac{4}{\sqrt{5}-1} = \sqrt{5} + 1$$

$$\text{Let } \text{cosec } 18^\circ = x = \sqrt{5} + 1$$

$$\Rightarrow x - 1 = \sqrt{5}$$

Squaring both sides, we get

$$x^2 - 2x + 1 = 5$$

$$\Rightarrow x^2 - 2x - 4 = 0.$$

Q76: The length of the latus rectum of a parabola, whose vertex and focus are on the positive x-axis at a distance R and S ($> R$) respectively from the origin, is ?

(A) $4(S + R)$

(B) $2(S + R)$

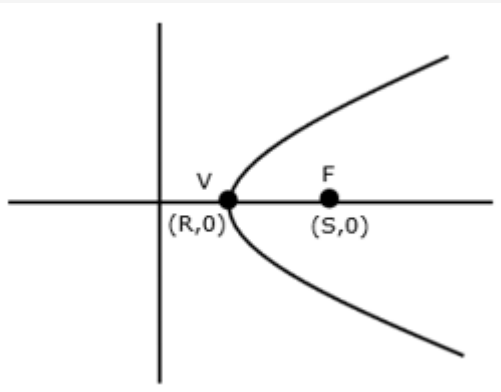
(C) $2(S - R)$

(D) $4(S - R)$

Q76: The length of the latus rectum of a parabola, whose vertex and focus are on the positive x-axis at a distance R and S ($> R$) respectively from the origin, is ?

- (A) $4(S + R)$
- (B) $2(S + R)$
- (C) $2(S - R)$
- (D) $4(S - R)$

Solution:



$V \rightarrow$ Vertex

$F \rightarrow$ Focus

$$VF = S - R.$$

Therefore, Length of latus rectum $= 4(S - R)$

Q77: If the following system of linear equations

$$2x + y + z = 5$$

$$x - y + z = 3$$

$$x + y + az = b$$

Has no solution, then ?

(A) $a \neq \frac{1}{3}, b = \frac{7}{3}$

(B) $a \neq -\frac{1}{3}, b = \frac{7}{3}$

(C) $a = \frac{1}{3}, b \neq \frac{7}{3}$

(D) $a = -\frac{1}{3}, b \neq \frac{7}{3}$

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(D) $a = -\frac{1}{3}, b \neq \frac{7}{3}$

Solution:

$$\text{Here } D = \begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & a \end{vmatrix} = 2(-a-1) - 1(a-1) + 1+1 = 1-3a$$

$$D_3 = \begin{vmatrix} 2 & 1 & 5 \\ 1 & -1 & 3 \\ 1 & 1 & b \end{vmatrix} = 2(-b-3) - 1(b-3) + 5(1+1) = 7-3b$$

For $a = \frac{1}{3}, b \neq \frac{7}{3}$, system has no solutions.

Q78: Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left| 2\vec{a} + 3\vec{b} \right| = \left| 3\vec{a} + \vec{b} \right|$ and the angle between $\vec{a}$ and $\vec{b}$ is 60° . If $\frac{1}{8}\vec{a}$ is a unit vector, then $\left| \vec{b} \right|$ is equal to ?

(A) 5

(B) 8

(C) 4

(D) 6

Q78: Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left| 2\vec{a} + 3\vec{b} \right| = \left| 3\vec{a} + \vec{b} \right|$ and the angle between $\vec{a}$ and $\vec{b}$ is 60° . If $\frac{1}{8}\vec{a}$ is a unit vector, then $\left| \vec{b} \right|$ is equal to ?

(A) 5

(B) 8

(C) 4

(D) 6

Solution:

$$|3\vec{a} + \vec{b}|^2 = |2\vec{a} + 3\vec{b}|^2$$

$$(3\vec{a} + \vec{b}) \cdot (3\vec{a} + \vec{b}) = (2\vec{a} + 3\vec{b}) \cdot (2\vec{a} + 3\vec{b})$$

$$9\vec{a} \cdot \vec{a} + 6\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b} = 4\vec{a} \cdot \vec{a} + 12\vec{a} \cdot \vec{b} + 9\vec{b} \cdot \vec{b}$$

$$5|\vec{a}|^2 - 6\vec{a} \cdot \vec{b} = 8|\vec{b}|^2$$

$$5(8)^2 - 6 \cdot 8 \cdot |\vec{b}| \cos 60^\circ = 8|\vec{b}|^2 \quad \left(\begin{array}{l} \because \frac{1}{8}|\vec{a}| = 1 \\ \Rightarrow |\vec{a}| = 8 \end{array} \right)$$

$$40 - 3|\vec{b}| = |\vec{b}|^2$$

$$\Rightarrow |\vec{b}|^2 + 3|\vec{b}| - 40 = 0$$

$$|\vec{b}| = -8, |\vec{b}| = 5$$

Q79: If $\frac{dy}{dx} = \frac{2^{x+y} - 2^x}{2^y}$, $y(0) = 1$, then $y(1)$ is equal to ?

(A) $\log_2 (2 + e)$

(B) $\log_2 (1 + e)$

(C) $\log_2 (2e)$

(D) $\log_2 (1 + e^2)$

Q79: If $\frac{dy}{dx} = \frac{2^{x+y}-2^x}{2^y}$, $y(0) = 1$, then $y(1)$ is equal to ?

(A) $\log_2(2 + e)$

(B) $\log_2(1 + e)$

(C) $\log_2(2e)$

(D) $\log_2(1 + e^2)$

Solution:

$$\frac{dy}{dx} = \frac{2^x 2^y - 2^x}{2^y}$$

$$2^y \frac{dy}{dx} = 2^x (2^y - 1)$$

$$\int \frac{2^y}{2^y - 1} dy = \int 2^x dx$$

$$\frac{\ln(2^y - 1)}{\ln 2} = \frac{2^x}{\ln 2} + C$$

$$\Rightarrow \log_2(2^y - 1) = 2^x \log_2 e + C$$

$$\because y(0) = 1 \Rightarrow 0 = \log_2 e + C$$

$$C = -\log_2 e$$

$$\Rightarrow \log_2(2^y - 1) = (2^x - 1) \log_2 e$$

$$\text{Put } x = 1, \log_2(2^y - 1) = \log_2 e$$

$$2^y = e + 1$$

$$y = \log_2(e + 1)$$

Q80: Let $*$, $\square \in \{\wedge, \vee\}$ be such that the Boolean expression $(p * \sim q) \Rightarrow (p \square q)$ is a tautology. Then :

(A) $* = \vee, \square = \wedge$

(B) $* = \wedge, \square = \vee$

(C) $* = \wedge, \square = \wedge$

(D) $* = \vee, \square = \vee$

Q80: Let $*$, $\square \in \{\wedge, \vee\}$ be such that the Boolean expression $(p * \sim q) \Rightarrow (p \square q)$ is a tautology. Then :

(A) $*$ = $\vee$, $\square$ = $\wedge$

(B) $*$ = $\wedge$, $\square$ = $\vee$

(C) $*$ = $\wedge$, $\square$ = $\wedge$

(D) $*$ = $\vee$, $\square$ = $\vee$

Solution:

$(p \wedge \sim q) \rightarrow (p \vee q)$ is tautology

p	q	$\sim q$	$p \wedge \sim q$	$p \vee q$	$(p \wedge \sim q) \rightarrow (p \vee q)$
T	T	F	F	T	T
T	F	T	T	T	T
F	T	F	F	T	T
F	F	T	F	F	T

Q81: If $x\phi(x) = \int_5^x (3t^2 - 2\phi'(t)) dt$, $x > -2$, and $\phi(0) = 4$, then $\phi(2)$ is

_____?

Q81: If $x\phi(x) = \int_5^x (3t^2 - 2\phi'(t)) dt$, $x > -2$, and $\phi(0) = 4$, then $\phi(2)$ is _____?

4.00

Solution:

$$x\phi(x) = \int_5^x (3t^2 - 2\phi'(t)) dt$$

$$x\phi(x) = x^3 - 125 - 2[\phi(x) - \phi(5)]$$

$$x\phi(x) = x^3 - 125 - 2\phi(x) + 2\phi(5)$$

$$\phi(0) = 4 \Rightarrow \phi(5) = -\frac{133}{2}$$

$$\phi(x) = \frac{x^3+8}{x+2}$$

$$\phi(2) = 4.$$

Q82: The square of the distance of the point of intersection of the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6}$ and the plane $2x - y + z = 6$ from the point $(-1, -1, 2)$ is _____ ?

Q82: The square of the distance of the point of intersection of the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6}$ and the plane $2x - y + z = 6$ from the point $(-1, -1, 2)$ is _____ ?

61.00

Solution:

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6} = \lambda$$

$$x = 2\lambda + 1, y = 3\lambda + 2, z = 6\lambda - 1$$

For point of intersection of line and plane

$$2(2\lambda + 1) - (3\lambda + 2) + (6\lambda - 1) = 6$$

$$7\lambda = 7 \Rightarrow \lambda = 1$$

point : $(3, 5, 5)$

$$\begin{aligned} (\text{distance})^2 &= (3 + 1)^2 + (5 + 1)^2 + (5 - 2)^2 \\ &= 16 + 36 + 9 = 61 \end{aligned}$$

Q83: If $\left(\frac{3^6}{4^4}\right) k$ is the term independent of x in the binomial expansion of $\left(\frac{x}{4} - \frac{12}{x^2}\right)^{12}$, then k is equal to _____ ?

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55.00

Solution:

$$\left(\frac{x}{4} - \frac{12}{x^2}\right)^{12}$$

$$T_{r+1} = (-1)^r \cdot {}^{12}C_r \left(\frac{x}{4}\right)^{12-r} \left(\frac{12}{x^2}\right)^r$$

$$T_{r+1} = (-1)^r \cdot {}^{12}C_r \left(\frac{1}{4}\right)^{12-r} (12)^r \cdot (x)^{12-3r}$$

$$\text{Term independent of } x \Rightarrow 12 - 3r = 0 \Rightarrow r = 4$$

$$T_5 = (-1)^4 \cdot {}^{12}C_4 \left(\frac{1}{4}\right)^8 (12)^4 = \frac{3^6}{4^4} \cdot k$$

$$\Rightarrow k = 55.$$

Q84: An electric instrument consists of two units. Each unit must function independently for the instrument to operate. The probability that the first unit functions is 0.9 and that of the second unit is 0.8. The instrument is switched on and it fails to operate. If the probability that only the first unit failed and second unit is functioning is p , then $98p$ is equal to _____
?

Q84: An electric instrument consists of two units. Each unit must function independently for the instrument to operate. The probability that the first unit functions is 0.9 and that of the second unit is 0.8. The instrument is switched on and it fails to operate. If the probability that only the first unit failed and second unit is functioning is p , then $98p$ is equal to _____ ?

Solution:

I_1 = first unit is functioning

I_2 = second unit is functioning

$$p(I_1) = 0.9, p(I_2) = 0.8$$

$$p(\bar{I}_1) = 0.1, p(\bar{I}_2) = 0.2$$

$$p = \frac{0.8 \times 0.1}{0.1 \times 0.2 + 0.9 \times 0.2 + 0.1 \times 0.8} = \frac{8}{28}$$

$$98p = \frac{8}{28} \times 98 = 28.$$

Q85: A point z moves in the complex plane such that $\arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{4}$, then the minimum value of $|z - 9\sqrt{2} - 2i|^2$ is equal to _____ ?

Q85: A point z moves in the complex plane such that $\arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{4}$, then the minimum value of $|z - 9\sqrt{2} - 2i|^2$ is equal to _____ ?

Solution:

Let $z = x + iy$

$$\arg\left(\frac{x-2+iy}{x+2+iy}\right) = \frac{\pi}{4}$$

$$\arg(x-2+iy) - \arg(x+2+iy) = \frac{\pi}{4}$$

$$\tan^{-1}\left(\frac{y}{x-2}\right) - \tan^{-1}\left(\frac{y}{x+2}\right) = \frac{\pi}{4}$$

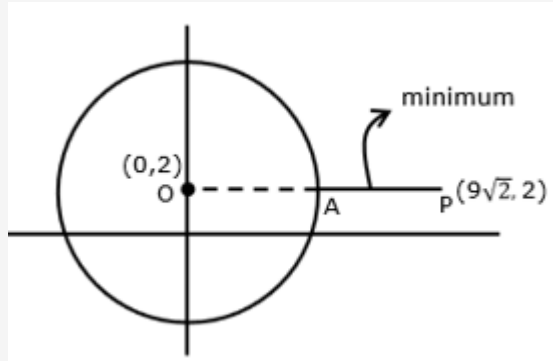
$$\frac{\frac{y}{x-2} - \frac{y}{x+2}}{1 + \left(\frac{y}{x-2}\right) \cdot \left(\frac{y}{x+2}\right)} = \tan \frac{\pi}{4} = 1$$

$$\frac{xy+2y-xy+2y}{x^2-4+y^2} = 1$$

$$4y = x^2 - 4 + y^2$$

$$x^2 + y^2 - 4y - 4 = 0$$

Locus is a circle with centre $(0, 2)$ and radius $= 2\sqrt{2}$



$$\text{min. value} = (AP)^2 = (OP - OA)^2$$

$$= (9\sqrt{2} - 2\sqrt{2})^2$$

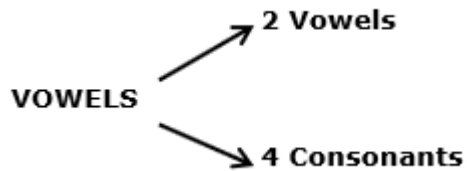
$$= (7\sqrt{2})^2 = 98.$$

Q86: The number of six letter words (with or without meaning), formed using all the letters of the word 'VOWELS', so that all the consonants never come together, is _____ ?

Q86: The number of six letter words (with or without meaning), formed using all the letters of the word 'VOWELS', so that all the consonants never come together, is _____ ?

576.00

Solution:



All Consonants should not be together
= Total – All consonants together,
= $6! - 3!4! = 576$.

Q87: The mean of 10 numbers $7 \times 8, 10 \times 10, 13 \times 12, 16 \times 14, \dots$ is

?

Q87: The mean of 10 numbers $7 \times 8, 10 \times 10, 13 \times 12, 16 \times 14, \dots$ is

?

398.00

Solution:

$$7 \times 8, 10 \times 10, 13 \times 12, 16 \times 14, \dots$$

$$T_n = (3n + 4)(2n + 6) = 2(3n + 4)(n + 3)$$

$$= 2(3n^2 + 13n + 12) = 6n^2 + 26n + 24$$

$$S_{10} = \sum_{n=1}^{10} T_n = 6 \sum_{n=1}^{10} n^2 + 26 \sum_{n=1}^{10} n + 24 \sum_{n=1}^{10} 1$$

$$= \frac{6(10 \times 11 \times 21)}{6} + 26 \times \frac{10 \times 11}{2} + 24 \times 10$$

$$= 10 \times 11(21 + 13) + 240$$

$$= 3980$$

$$\text{Mean} = \frac{S_{10}}{10} = \frac{3980}{10} = 398.$$

Q88: If ' R ' is the least value of ' a ' such that the function $f(x) = x^2 + ax + 1$ is increasing on $[1, 2]$ and ' S ' is the greatest value of ' a ' such that the function $f(x) = x^2 + ax + 1$ is decreasing on $[1, 2]$, then the value of $|R - S|$ is _____ ?

Q88: If ' R ' is the least value of ' a ' such that the function $f(x) = x^2 + ax + 1$ is increasing on $[1, 2]$ and ' S ' is the greatest value of ' a ' such that the function $f(x) = x^2 + ax + 1$ is decreasing on $[1, 2]$, then the value of $|R - S|$ is _____ ?

2.00

Solution:

$$f(x) = x^2 + ax + 1$$

$$f'(x) = 2x + a$$

when $f(x)$ is increasing on $[1, 2]$

$$2x + a \geq 0 \forall x \in [1, 2]$$

$$a \geq -2x \forall x \in [1, 2]$$

$$R = -4$$

when $f(x)$ is decreasing on $[1, 2]$

$$2x + a \leq 0 \forall x \in [1, 2]$$

$$a \leq -2x \forall x \in [1, 2]$$

$$S = -2$$

$$|R - S| = |-4 + 2| = 2.$$

Q89: Let $[t]$ denote the greatest integer $\leq t$. Then the value of $8 \cdot \int_{-\frac{1}{2}}^1 ([2x] + |x|) dx$ is

_____?

Q89: Let $[t]$ denote the greatest integer $\leq t$. Then the value of $8 \cdot \int_{-\frac{1}{2}}^1 ([2x] + |x|) dx$ is

?

5.00

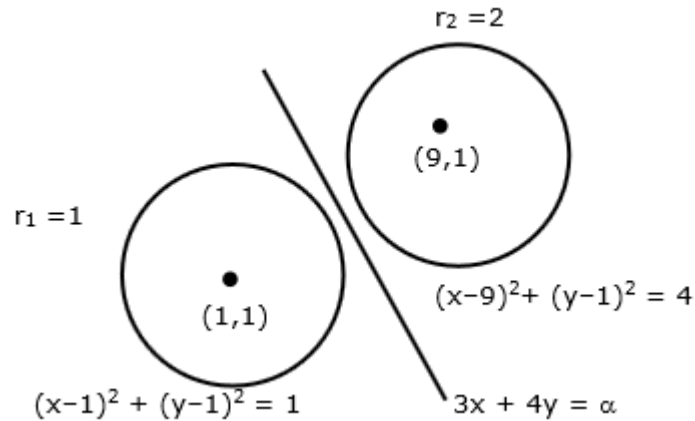
Solution:

$$\begin{aligned}
 I &= \int_{-\frac{1}{2}}^0 ([2x] + |x|) dx + \int_0^{\frac{1}{2}} ([2x] + |x|) dx + \int_{\frac{1}{2}}^1 ([2x] + |x|) dx \\
 &= \int_{-\frac{1}{2}}^0 (-1 - x) dx + \int_0^{\frac{1}{2}} (x) dx + \int_{\frac{1}{2}}^1 (1 + x) dx \\
 &= \left[-x - \frac{x^2}{2} \right]_{-\frac{1}{2}}^0 + \left[\frac{x^2}{2} \right]_0^{\frac{1}{2}} + \left[x + \frac{x^2}{2} \right]_{\frac{1}{2}}^1 \\
 &= \frac{5}{8} \\
 \Rightarrow 8I &= 5
 \end{aligned}$$

Q90: If the variable line $3x + 4y = \alpha$ lies between the two circles $(x - 1)^2 + (y - 1)^2 = 1$ and $(x - 9)^2 + (y - 1)^2 = 4$, without intercepting a chord on either circle, then the sum of all the integral values of α is _____ ?

Q90: If the variable line $3x + 4y = \alpha$ lies between the two circles $(x - 1)^2 + (y - 1)^2 = 1$ and $(x - 9)^2 + (y - 1)^2 = 4$, without intercepting a chord on either circle, then the sum of all the integral values of α is _____ ?

Solution:



Both centres should lie on either side of the line as well as line can be tangent to circle.

$$(3 + 4 - \alpha) \cdot (27 + 4 - \alpha) < 0$$

$$(7 - \alpha) \cdot (31 - \alpha) < 0 \Rightarrow \alpha \in (7, 31) \quad \dots (1)$$

d_1 = distance of (1,1) from line

d_2 = distance of (9,1) from line

$$d_1 \geq r_1 \Rightarrow \left| \frac{7-\alpha}{5} \right| \geq 1 \Rightarrow \alpha \in (-\infty, 2] \cup [12, \infty) \quad \dots (2)$$

$$d_2 \geq r_2 \Rightarrow \left| \frac{31-\alpha}{5} \right| \geq 2 \Rightarrow \alpha \in (-\infty, 21] \cup [41, \infty) \quad \dots (3)$$

$$(1) \cap (2) \cap (3) \Rightarrow \alpha \in [12, 21]$$

Sum of integers = 165